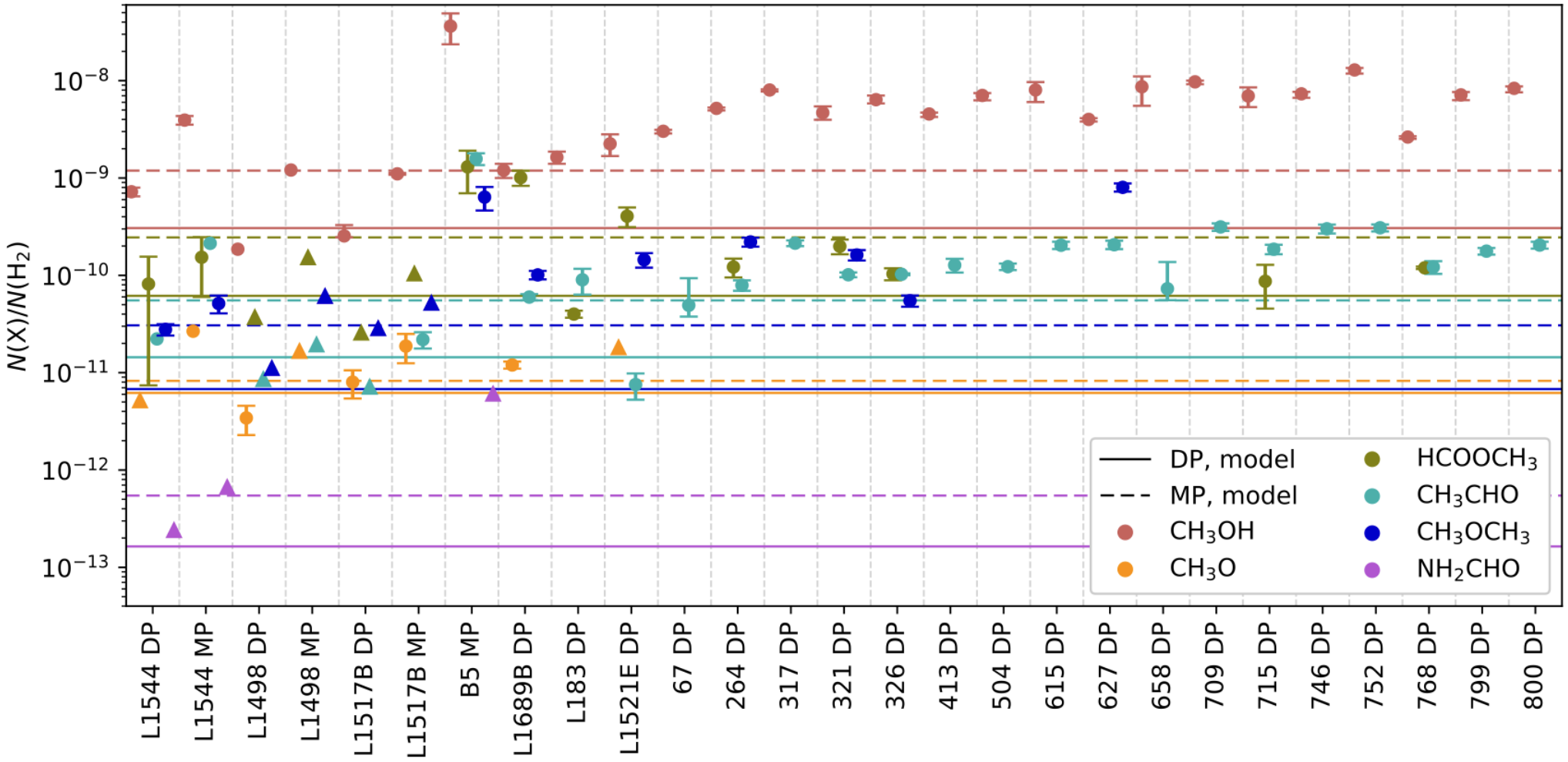
Impact of Cosmic Rays

Le N. Tram
Leiden University/Observatory – The Netherlands

UCLCHEM Workshop – Bonn 2026

Gas-phase COMs in Cold Environments

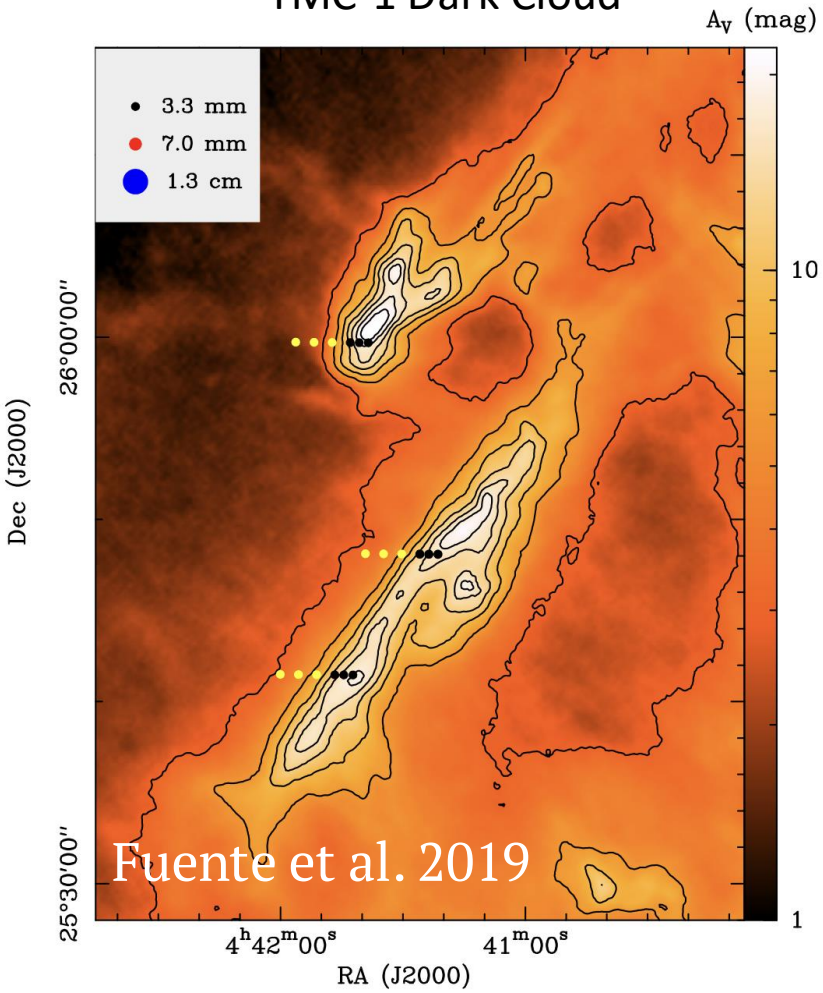
Pre-stellar cores



Borshcheva et al. 2025

Gas-phase COMs in Cold Environments

TMC-1 Dark Cloud



Fuente et al. 2019

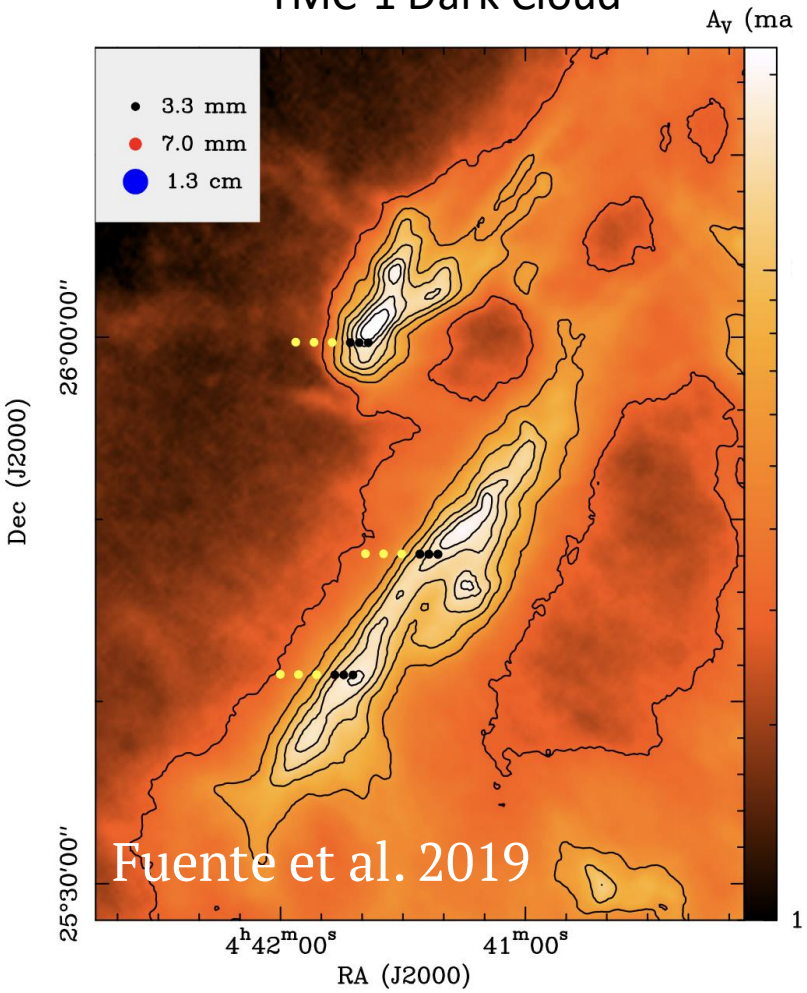
Table 1. Fractional abundances relative to $N = 2N(H_2)$ for species detected in TMC-1. – Indicates species is detected but no other information available. < Indicates species was searched for without success.

Species	Fractional abundance $N(x)/2N(H_2)$	Reference
H_2	0.5	Guélin *et al.* (1982a)
CH	$\sim 10^{-8}$	Brown (1981)
CO	3×10^{-5}	Allen & Knapp (1978)
CN	$\gtrsim 4 \times 10^{-9}$	Churchwell & Bieging (1983)
CS	–	Linke & Goldsmith (1980)
OH	$\sim 3 \times 10^{-8}$	Brown (1981)
SO	2.5×10^{-9}	Rydbeck *et al.* (1980)
HCN	6×10^{-9}	Irvine & Hjalmarson (1983)
HNC	4×10^{-9}	Irvine & Hjalmarson (1983)
C_2H	4.2×10^{-9}	Wootten *et al.* (1980)
HCO^+	$\sim 8 \times 10^{-9}$	Guélin *et al.* (1982a)
N_2H^+	–	Turner & Zuckerman (1978)
NH_3	$\sim 5 \times 10^{-8}$	Tölle *et al.* (1981)
H_2CO	$\sim 6 \times 10^{-9}$	Brown (1981)
HNCO	$\sim 9 \times 10^{-11}$	Brown (1981)
C_3N	$\sim 4 \times 10^{-9}$	Friberg *et al.* (1980)
HC_3N	$(0.3-1) \times 10^{-8}$	Guélin *et al.* (1982b)
HC_5N	5×10^{-9}	Tölle *et al.* (1981)
HC_7N	1.25×10^{-9}	Tölle *et al.* (1981)
HC_9N	1.5×10^{-10}	Winnewisser & Walmsley (1979)
CH_3CCH	$(2-4) \times 10^{-9}$	Irvine *et al.* (1981)
C_4H	$(0.8-1.5) \times 10^{-8}$	Guélin *et al.* (1982b)
CH_3C_3N	2.5×10^{-10}	Broten *et al.* (1983)
CH_3C_4H	$\sim 10^{-8}$	Avery (1983)
HCS^+	–	Irvine *et al.* (1983)
CH_3CN	3.4×10^{-10}	Matthews & Sears (1983a)
CH_2CHCN	1.5×10^{-10}	Matthews & Sears (1983b)
SO_2	<	Irvine *et al.* (1983)

Millar & Freeman 1984

Gas-phase COMs in Cold Environments

TMC-1 Dark Cloud



Fuente et al. 2019

A&A 673, A34 (2023)
https://doi.org/10.1051/0004-6361/202346076
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Astronomy & Astrophysics

Detection of ethanol, acetone, and propanal in TMC-1
New O-bearing complex organics in cold sources★

M. Agúndez[1], J.-C. Loison[2], K. M. Hickson[2], V. Wakel
N. Marcelino[4,5], B. Tercero[4,5], P. de Vicente

A&A 671, L6 (2023)
https://doi.org/10.1051/0004-6361/202245732
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Astronomy & Astrophysics

LETTER TO THE EDITOR

Discovery of CH_3CHCO in TMC-1 with the QUIJOTE line survey★

R. Fuentetaja[1], C. Bermúdez[2], C. Cabezas[1], M. Agúndez[1], B. Tercero[3,4], N. Marcelino[3,4], J. R. Pardo[1], L. Margulès[5], R. A. Motiyenko[5], J.-C. Guillemin[6], P. de Vicente[3], and J. Cernicharo[1]

Astronomy & Astrophysics

A&A 649, L4 (2021)
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LETTER TO THE EDITOR

O-bearing complex organic molecules at the cyanopolyyne peak of TMC-1: Detection of C_2H_3CHO, C_2H_3OH, $HCOOCH_3$, and CH_3OCH_3★

M. Agúndez[1], N. Marcelino[1], B. Tercero[2,3], C. Cabezas[1], P. de Vicente[3], and J. Cernicharo[1]

THE ASTROPHYSICAL JOURNAL, 854:116 (11pp), 2018 February 20
https://doi.org/10.3847/1538-4357/aaa70c
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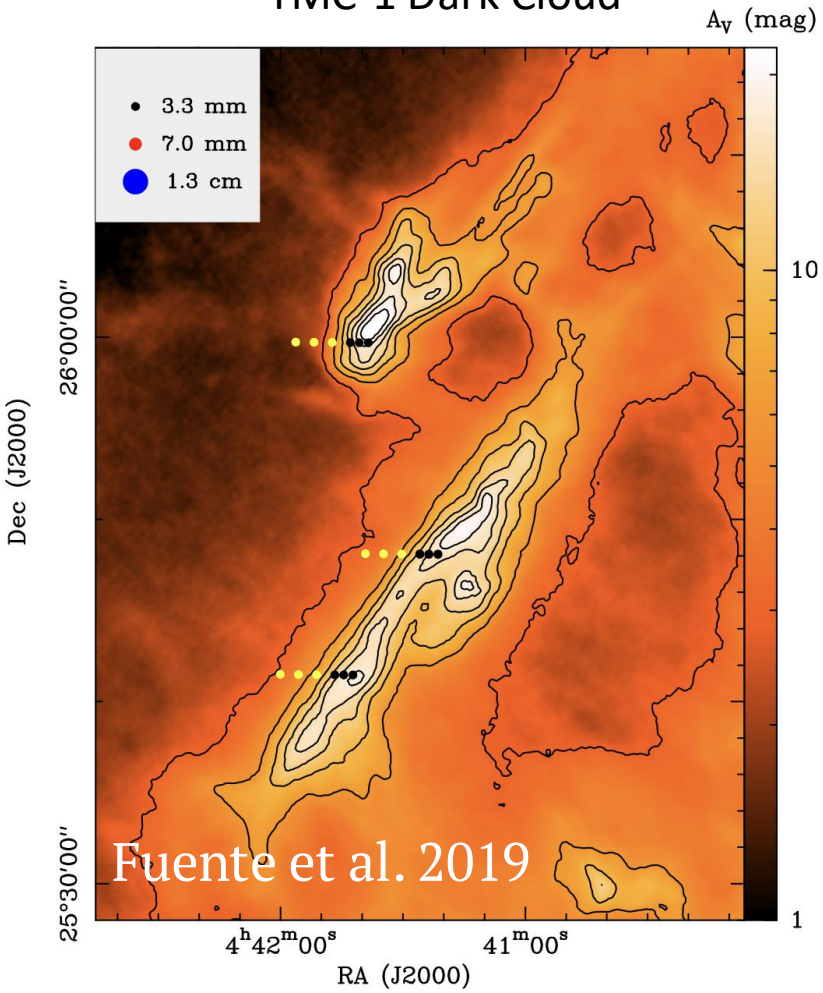
Complex Organic Molecules in Taurus Molecular Cloud-1

Tatsuya Soma[1], Nami Sakai[2], Yoshimasa Watanabe[1,3,4], and Satoshi Yamamoto[1]

Gas-phase COMs in Cold Environments

TMC-1 Dark Cloud



Fuente et al. 2019

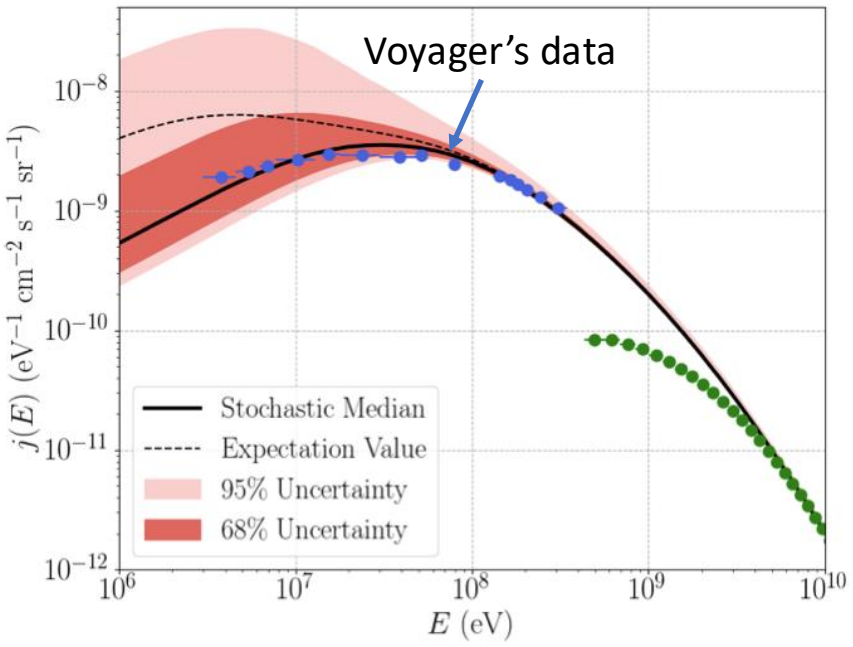
- External UV field is blocked!
- Absence of internal UV field!
- Where is the energy coming from?

Cosmic Rays as primary Driver

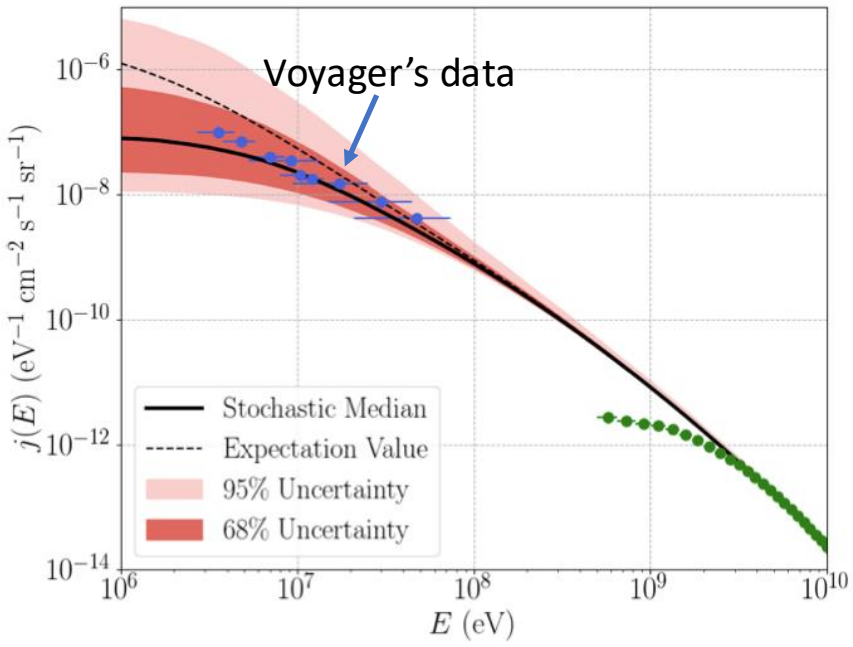
Cosmic Rays and their Origins?

- ❑ Cosmic rays are energetic particles (mostly proton, electron and nuclei)
- X Their paths cannot be traced back to the astrophysical origins
- ❑ Believed to be accelerated by shocks in Supernova Remnants
- ❑ Hot topic in Astrophysics/Astroparticles

Proton CR



electron CR



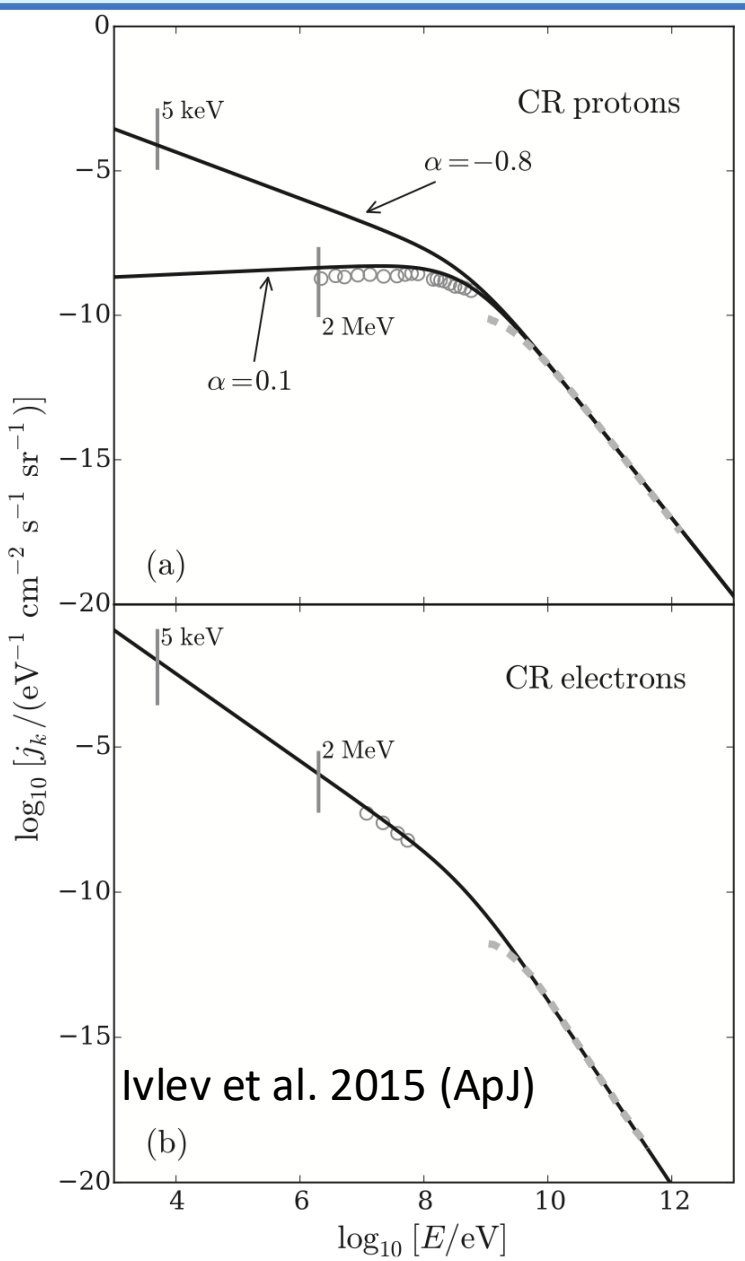
Phan et al. 2021 (PhysRevLett)

Cosmic Rays as primary Driver

- For proton CR
  - Model $\varkappa$
  - Model $\mathcal{L}$

- Flux of other ion nuclei heavier than proton nucleus
  - $j_k(E) = [k] j_p(E)$

$[k]$ abundance of the nucleus k



Ivlev et al. 2015 (ApJ)

Fundamental Effects of Cosmic Rays

- **CRs impact on gas**
  - Ionising gas
  - Directly heating up gas

- **CRs deposite energy on dust grains (deeper within the cloud)**
  - Directly Heating up grains (locally/stochastically)
  - Directly desorbing molecules from the grain's surface
  - Indirectly desorbing molecules from the grain's surface
    (Enhancing the diffusion rate of molecules on the grain's surface)
  - Directly ionizing and sputtering grains

- **CRs induce secondary UV radiation field (deeper within the cloud)**
  - Directly Heating up grains (globally)
  - Enabling photo-chemistry in dense gas

Cosmic-rays → Desorption and Surface Diffusion

- **Rate coefficient to directly desorb a molecule**

$$k_{\mathrm{CR}}(i) = f(a,\, T_{\mathrm{max}})\, k_{\mathrm{therm}}(i,\, T_{\mathrm{max}}),$$

- **The hopping timescale**

$$t_{\mathrm{hop}} = \nu_0^{-1} \exp\left(\frac{E_{\mathrm{b}}}{T_{\mathrm{CR}}}\right)$$

- **Rate coefficient for two molecules to diffuse and react**

$$k_{\mathrm{AB}} = f(a, T_{\mathrm{CR,max}})\kappa_{\mathrm{AB}}(k^{A}_{\mathrm{CR,hop}} + k^{B}_{\mathrm{CR,hop}})\frac{1}{N_{\mathrm{site}} n_{\mathrm{dust}}}$$

What does UCLCHEM include now?

- ✓ **Direct cosmic-ray desorption (CRD)**
  - ✗ Based on the effect of T_{CR} = 70 K on CO
  - ✗ Applied for all species

- ✗ **Cosmic-ray induced desorption (CRID)**
  - ➢ Mobilities of radical and other reactants on the surface

- ✓ **Secondary UV photons induced by cosmic-ray**
  - ✗ The secondary UV flux is 1/1000 times the ISRF's UV flux

- ✓ Effect of CR is controlled only by CR-ionization rate (CRIR)

Determinations of T_{CR} and $f(a,T_{CR})$

- **"Textboook", a value of T_{CR} = *70* K is used**
  (Hasegawa & Herbst, 1993), based on the work of Léger et al. 1985
  - If CRs deposit ~ 0.4 Mev thermal energy to grains → T_{CR} = 70 K

- **The factor $f(0.1\mu m, 70K) = 3.16 \times 10^{-19}$** $f(a, T_{max}) = \tau_{cool}/\tau_{heat}$

  τ_{heat}: Heating timescale
  τ_{cool}: Cooling timescale

- A series of Kalvans et al. modelled a "full spectrum" of T_{CR} for a given N_{gas} (Av)

- Sipila et al. (2021, 2025) adopted that spectrum to the chemical model (not only 70 K) to study the CR-induced direct desorption

Results from Kalvans et al. (using SRIM2013)

- **32 CR nuclei**
- **Grain composition**
  - Core: 60% Olivine ($MgFeSiO_4$) + 40% graphite (C)
  - Mantle: H_2O + CO + CO_2 ices

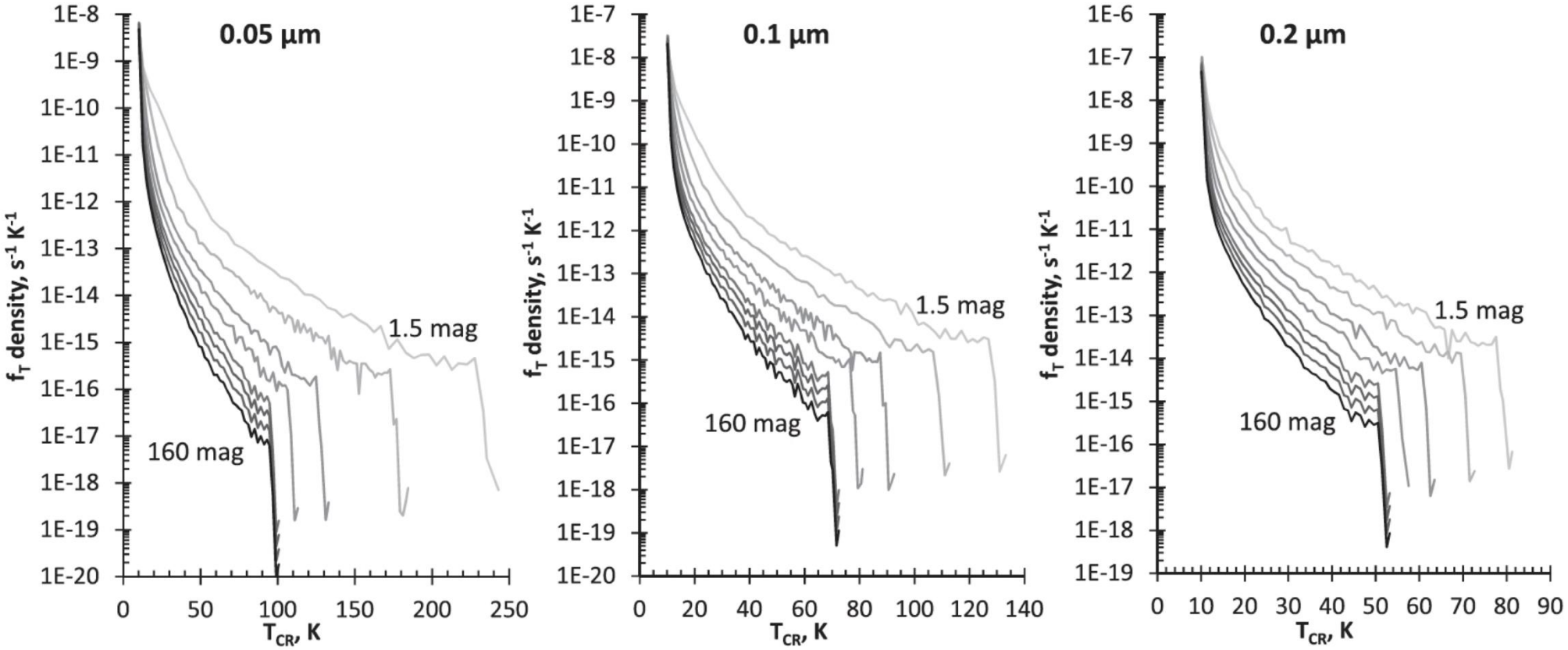
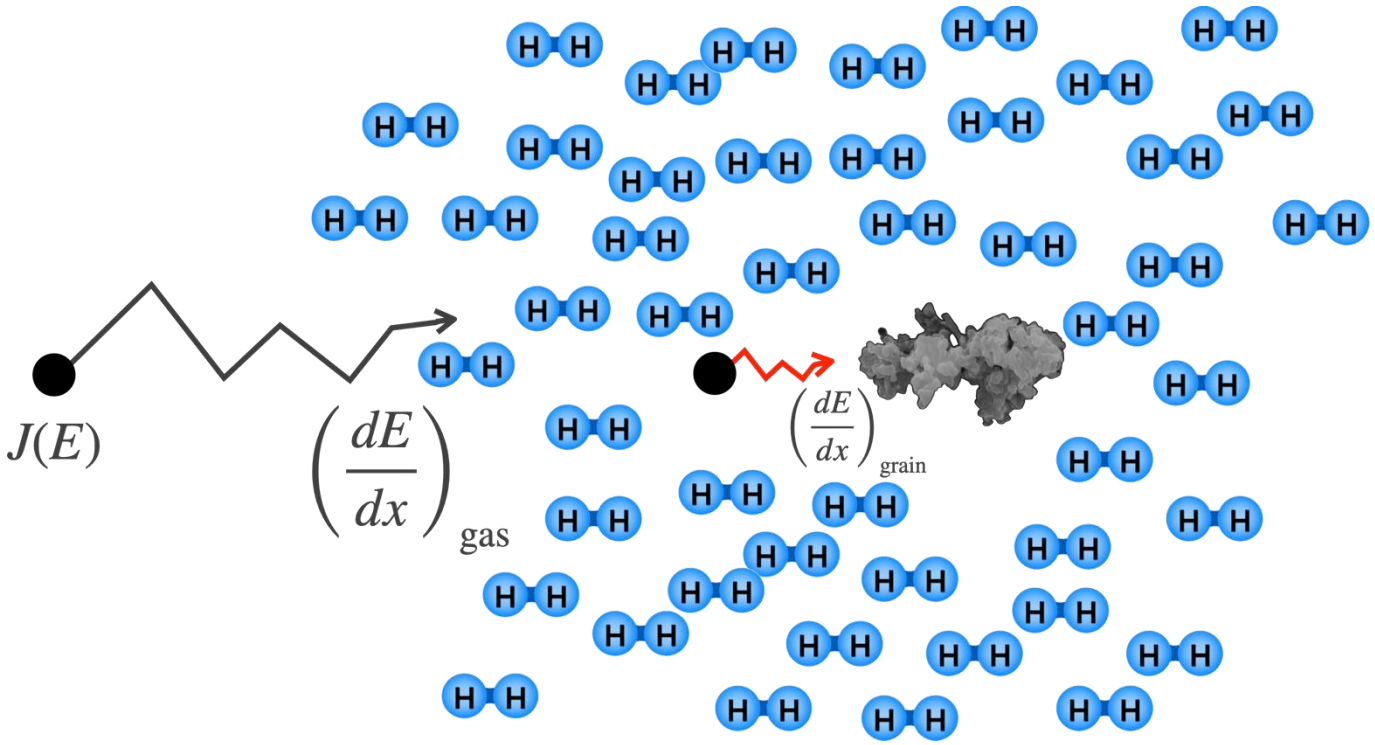


Figure 8. Temperature spectra for grains with core sizes 0.05, 0.1, and 0.2 μm hit by cosmic rays at A_V 1.5, 3.0, 6.0, 9.0, 20, 40, 80, and 160 mag, with $N_H/A_V = 2.20 \times 10^{21}$ cm^{-2}.

"In-house" model of CR-induced T_{CR} spectrum

- **Incident CR nucleous (i.e., projectile *P*)**
  - Kinetic energy E, charge $Z_P e$
  - Velocity βc (relativistic or non-relativistic)

- **Gas or Dust (i.e., target *T*)**
  - Kinetic energy E, charge $Z_T e$
  - Potential energy I_T



- **Grain receives 60% of energy deposited by CRs**

  Assumption: thermal energy

$$E_{\text{grain}} = \int_{10\,\text{K}}^{T_{\text{CR}}} C_V V_{\text{grain}} dT$$

- **Attenuated CR spectrum via CSDA approximation**

  (continuous slowing-down approximation)

$$j_k(E, N) = j_k(E_0) \frac{L_k(E_0)}{L_k(E)},$$

- **"Probability" of CRs-dust impact ← $j_k(E,N)$ + T_{CR}**

- **CRIR ← $j_k(E,N)$**

Loss function: fundamentals, $L_i(E)$

- **High-energy regime (relativistic CR)**
  - Bethe–Bloch theory

$$S_{\text{high-E,i}} = \frac{4\pi(Z_p e^2)^2}{m_e c^2 \beta^2} Z_{T,i} \left[\frac{1}{2} \ln \frac{2\gamma^2 m_e c^2 \beta^2 T_{\max}}{I_{T,i}^2} - \beta^2 - \frac{C}{Z_{T,i}} - \frac{\delta}{2} \right]$$

  - **Z_P correction** ← Anthony & Lanford, 1982
  - **C/Z_T shell correction** ← Bischsel 1972
  - **$\delta/2$ density correction** ← Sternheimer et al. 1984

- **Low-energy regime (non-relativistic CR)**
  - LLS theory (Lindhard & Scharff 1961)

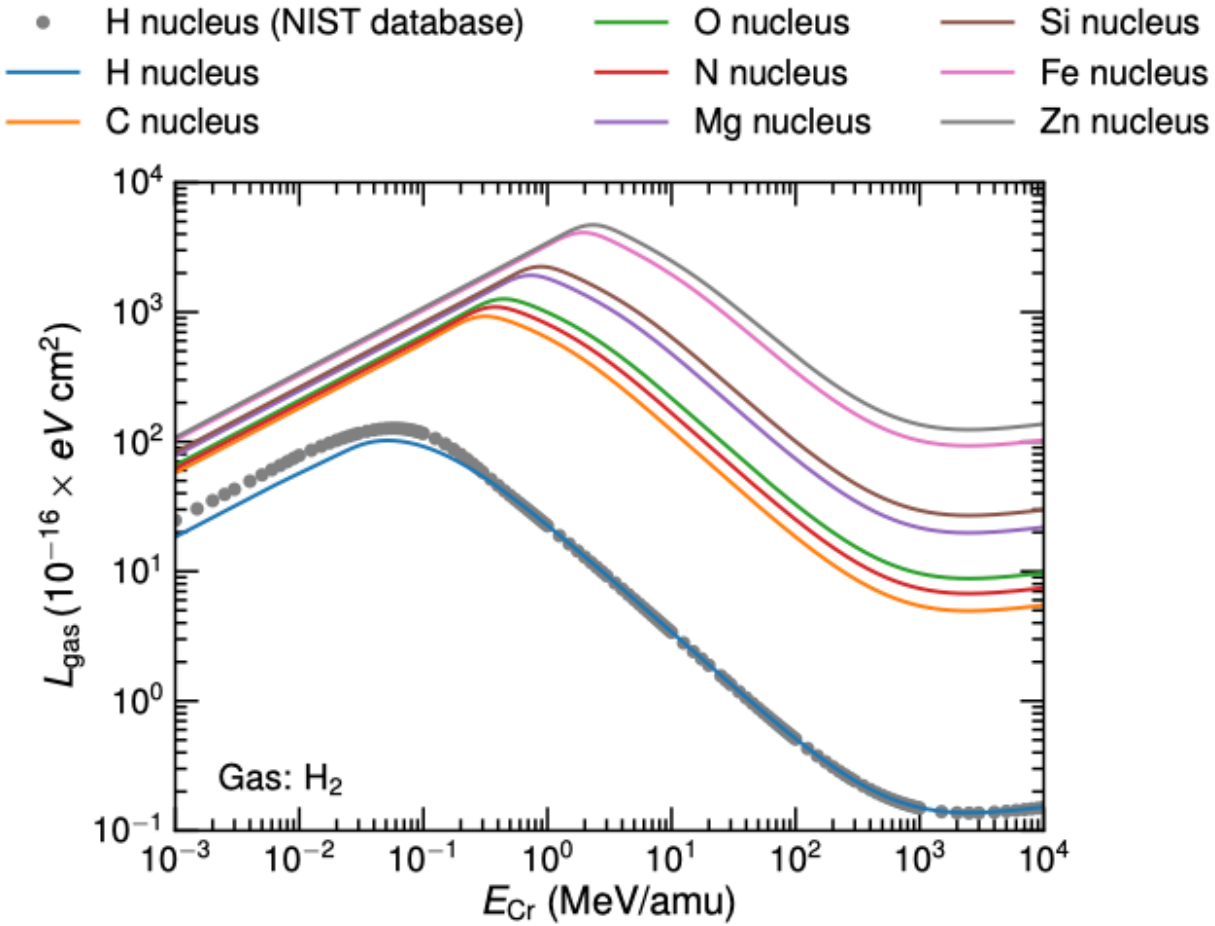
$$S_{\text{low-E,i}} = 8\pi e^2 a_0 Z_P^{1/6} \frac{Z_P Z_T}{\left(Z_P^{2/3} + Z_T^{2/3}\right)^{3/2}} \frac{\beta c}{v_{\text{Bohr}}}$$

- **Smooth function applied**

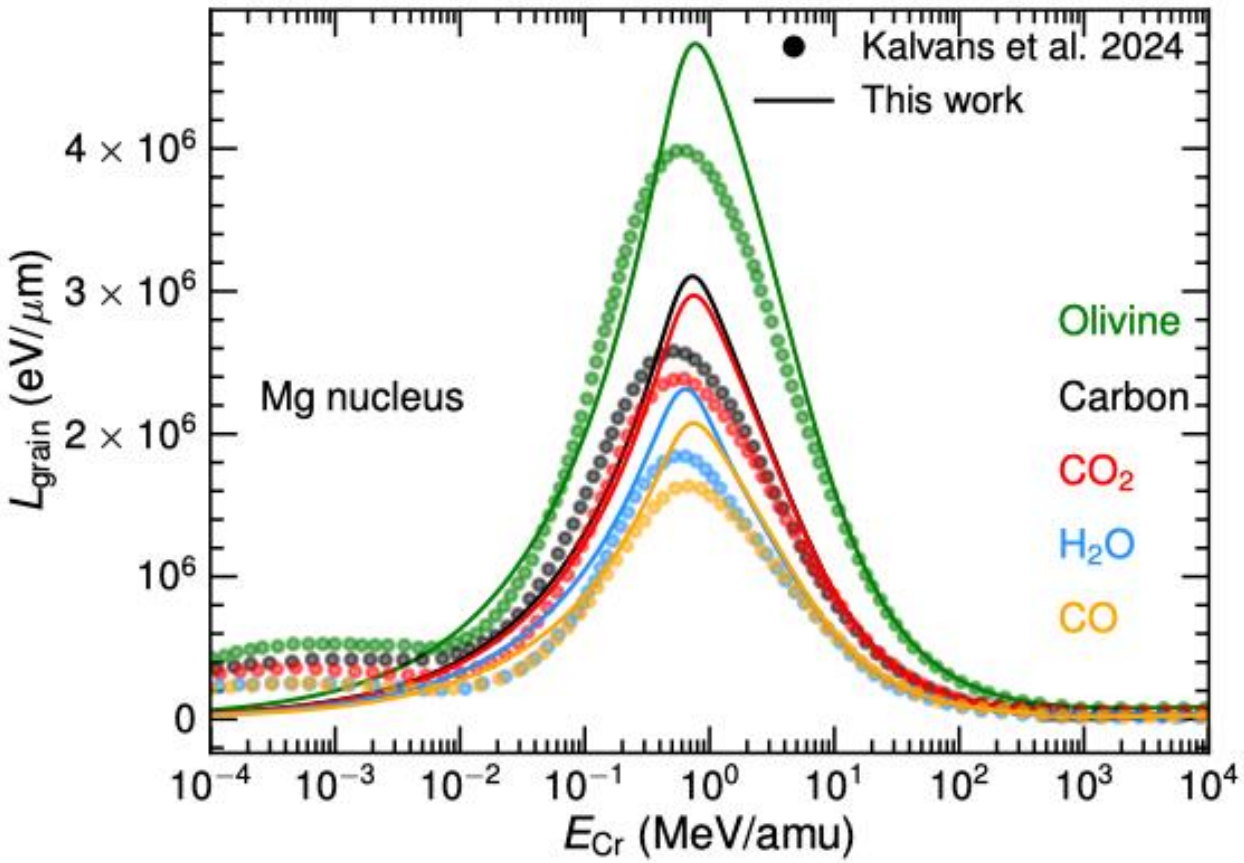
$$S_i = S_{\text{low-E,i}} + 0.5 \times (S_{\text{high-E,i}} - S_{\text{low-E,i}}) \times \left\{ 1 + \tanh \left[\frac{\ln(E/E_{\text{trans}})}{0.5} \right] \right\}$$

Loss function: benchmarks

❖ **Loss-energy of CRs on gas**

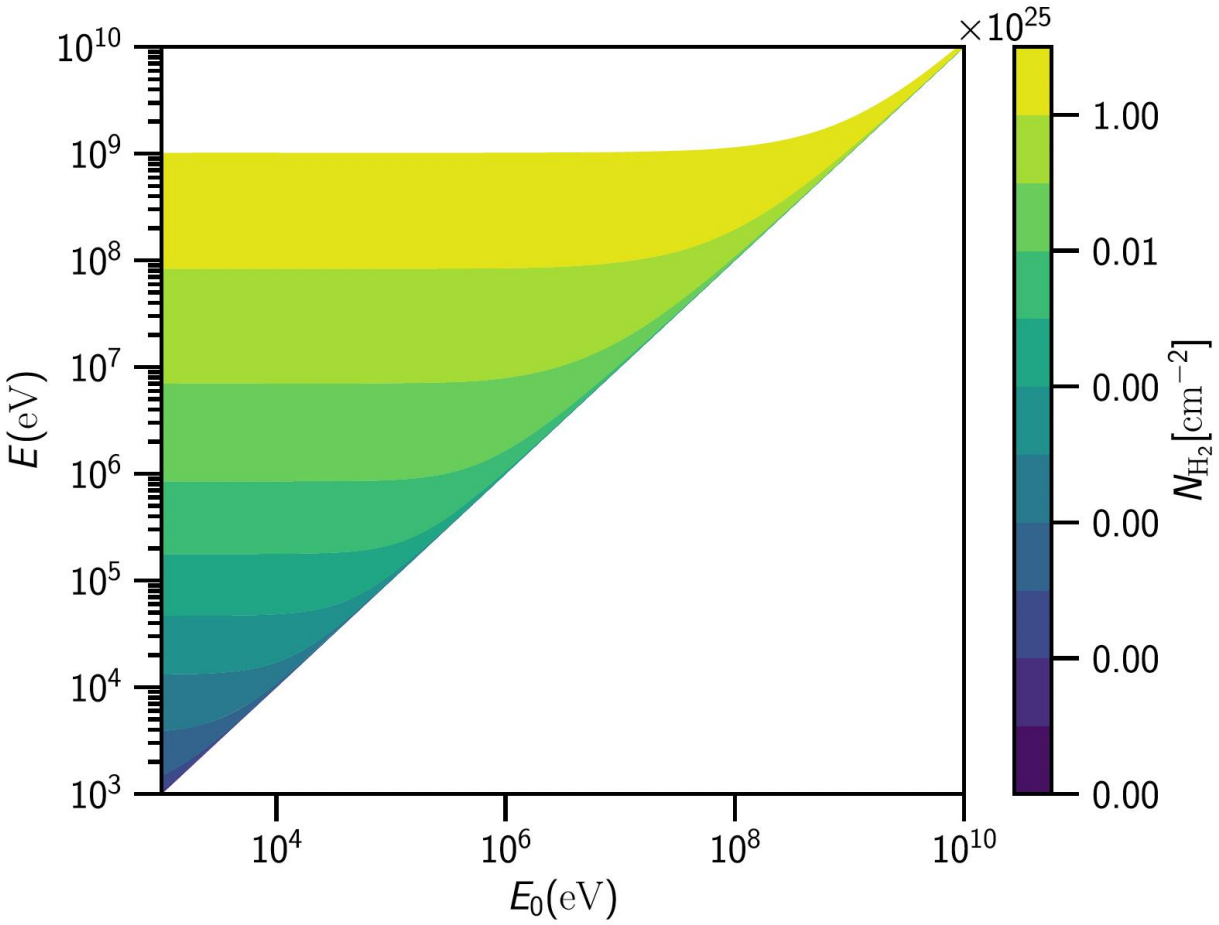


❖ **Loss-energy of CRs on dust**



E-E_0 relation: CSDA Receipe

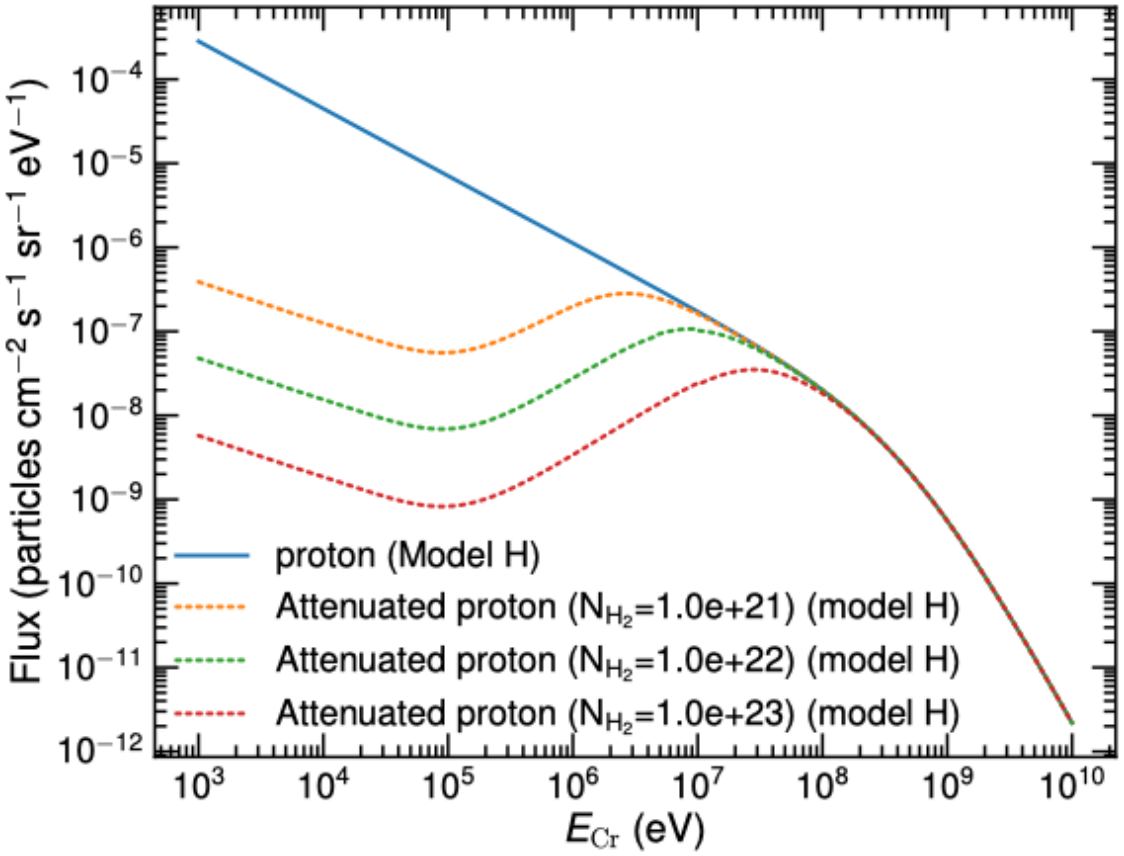
- Compute the gas density required to make a CR nucleous completely stop.
- Create a map of $N_{gas} = f(E,E_0)$
- At a given N_{gas}, find the E-E_0 relation (by fitting)



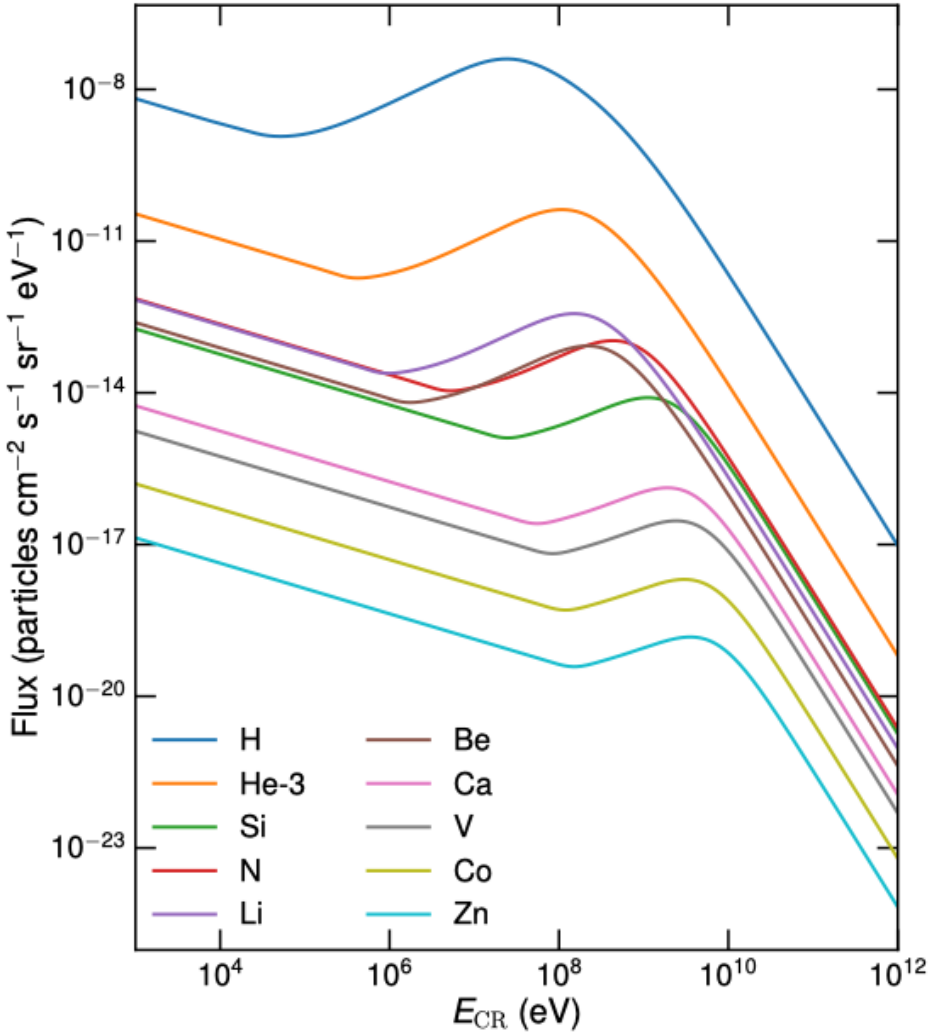
Hoang et al. 2025

Attenuated CR spectrum, $j_i(E,N)$

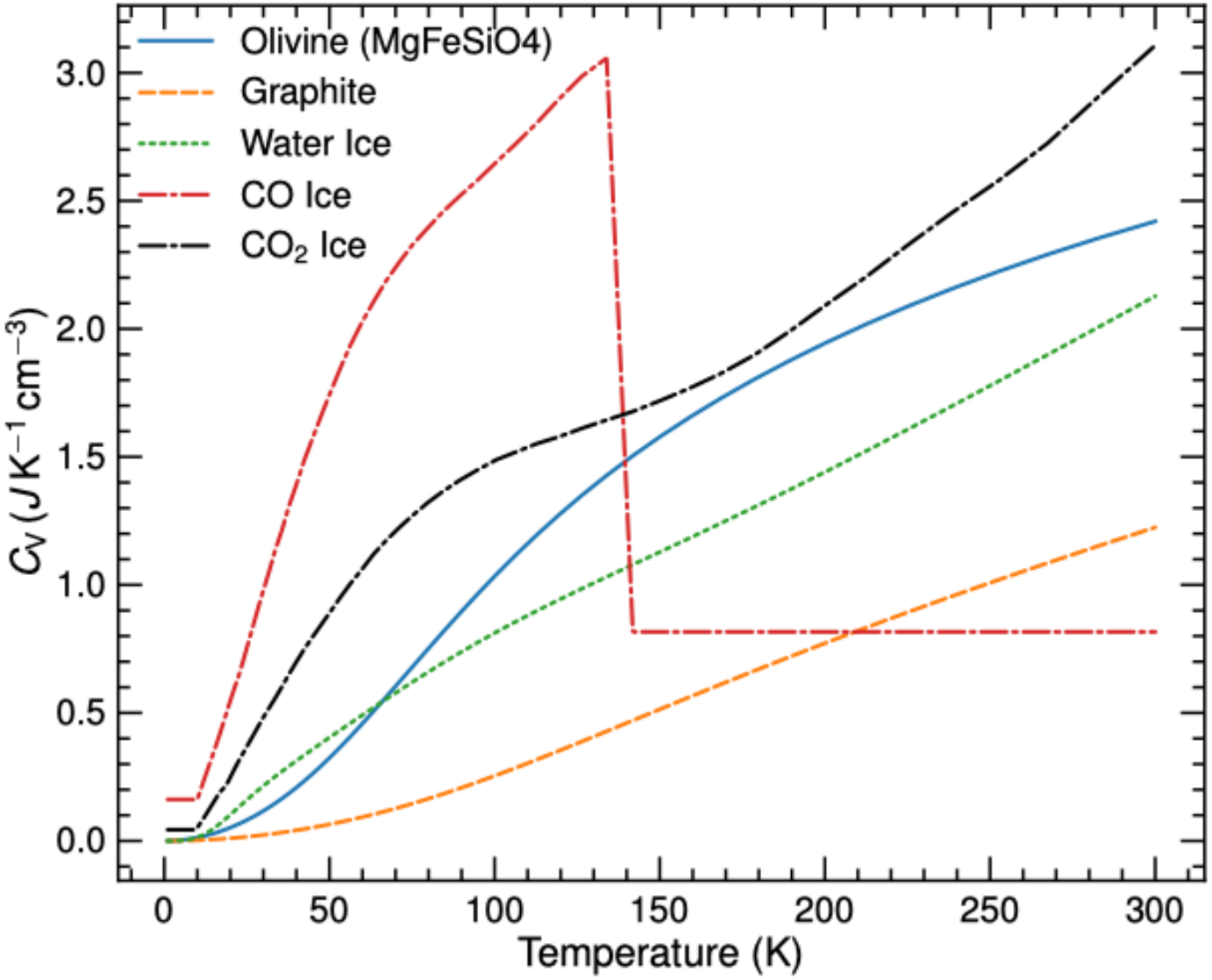
❖ **Light nucleus**



❖ **Heavy nuclei**



Heat Capacities of materials, C_V

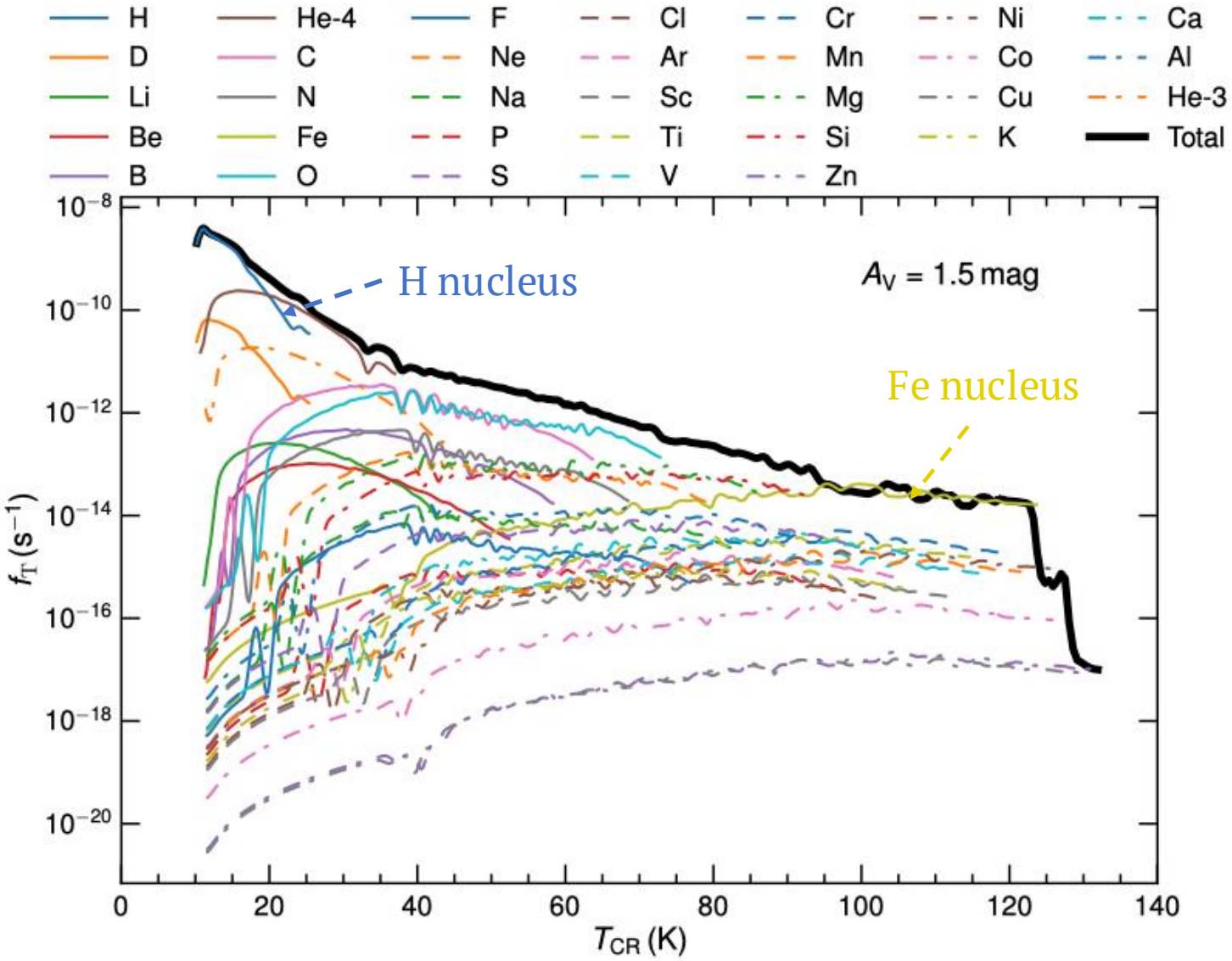


T_{CR} spectrum

Physical parameters					Value
Grain core radius a (μm)					0.1
Grain core composition					60% Olivine & 40% graphite
	$A_V < 3$	$3 \leq A_V < 6$	$6 \leq A_V < 9$	$9 \leq A_V < 20$	$A_V \geq 20$
Mantle thickness d (μm)	0	0.01	0.02	0.025	0.03
Water ice fraction on mantle f_{H_2O} (%)	0	78	72	60	50
CO ice fraction on mantle f_{CO} (%)	0	5	14	23	28
CO_2 ice fraction on mantle f_{CO_2}	0	17	14	17	22
Model acronyms					**Description**
Current UCLCHEM					Version of UCLCHEM model before this work
Update CR Desorptions					Direct desorption by CRs is updated
Update CR + CRI Desorptions					Direct and induced desorptions by CRs are updated
Update DEUVCR					UV desorptions by the ISRF and CR-induced fields are updated
Update CR + CRI + DEUVCR Desorptions					All updated
no heating/cooling					All updated with $T_{gas} \equiv T_{dust}$
with heating/cooling					All updated with $T_{gas} \neq T_{dust}$

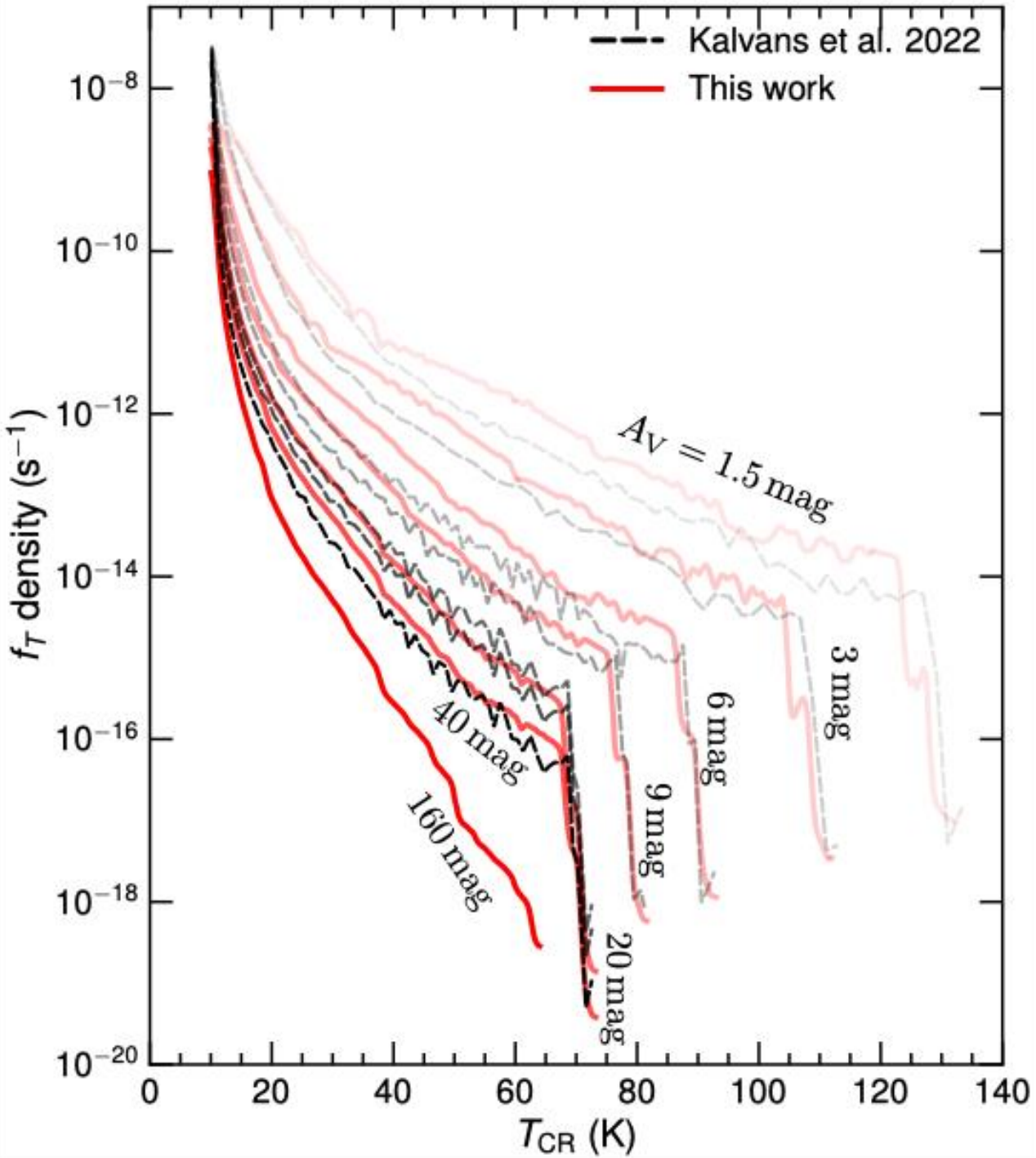
Tram et al. in prep.

T_{CR} spectrum



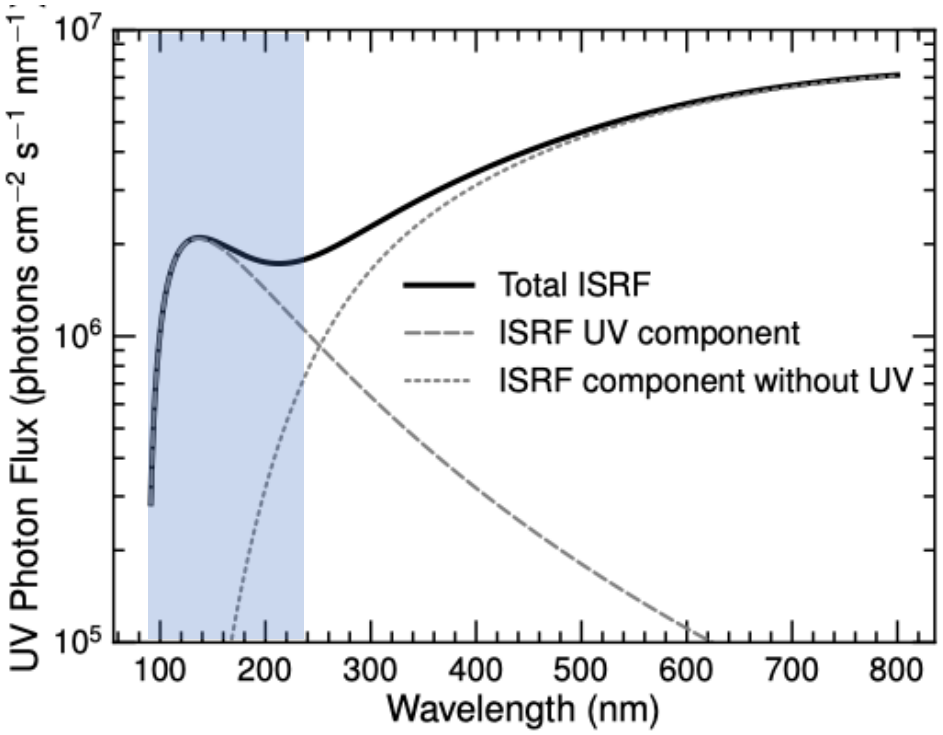
T_{CR} spectrum

- **For $A_v \leq 40$ mag** : A reasonable good agreement with Kalvans et al. is found.
- **For A_v>40 mag** : Our heavy nuclei are much more attenuated → lower frequencies.

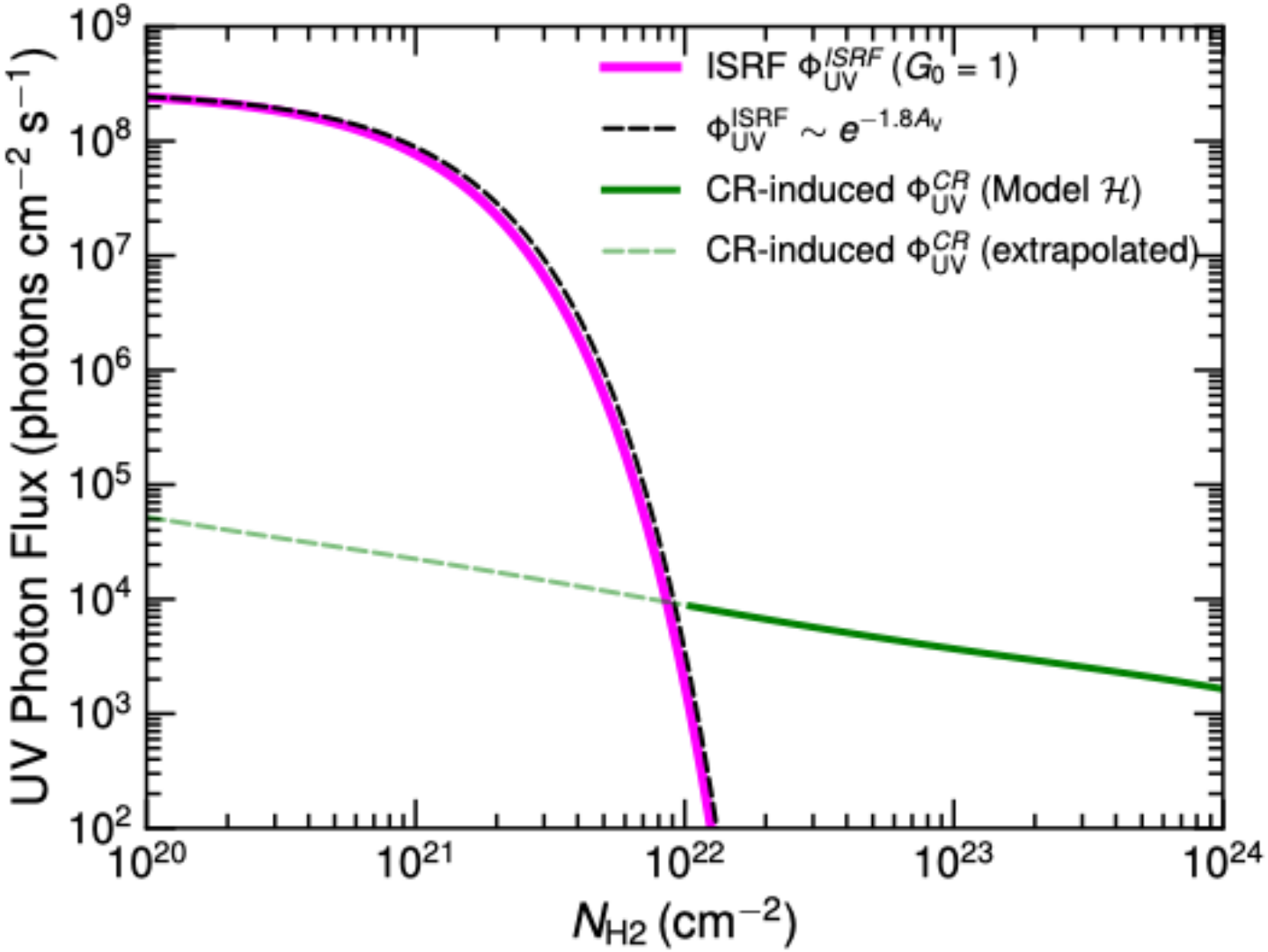


UV radiation fields: ISRF and CR-induced

Draine radiation field

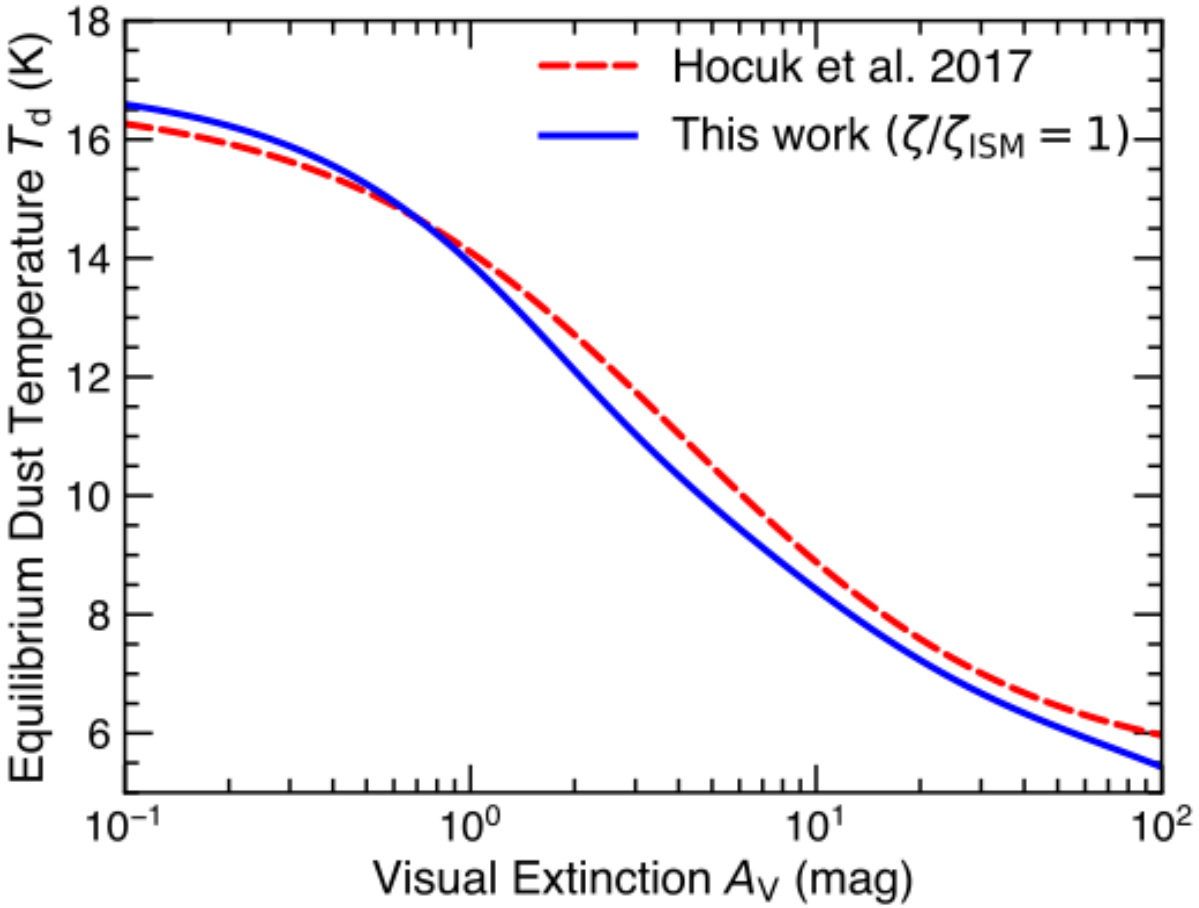


Integrated UV flux

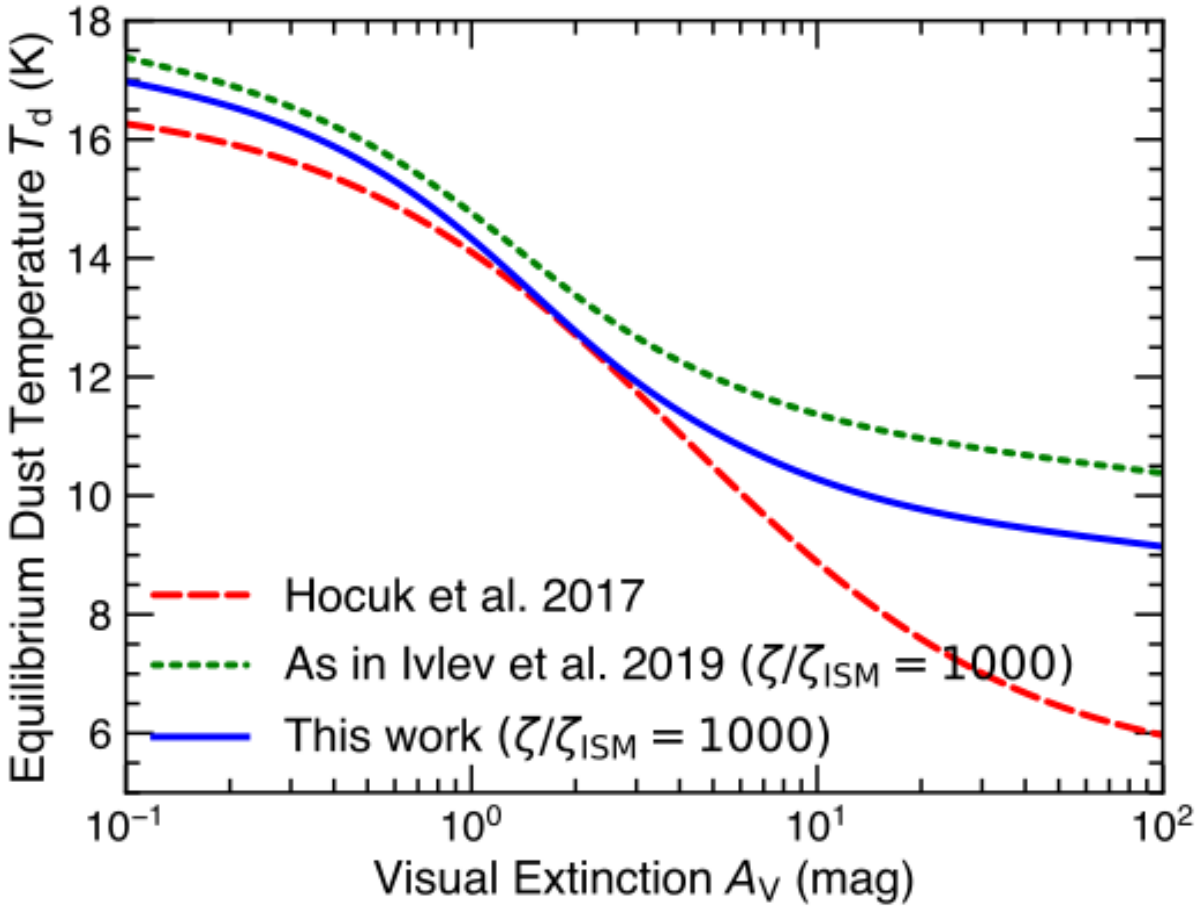


Cosmic-rays and Dust Temperature

Typical ISM CR flux

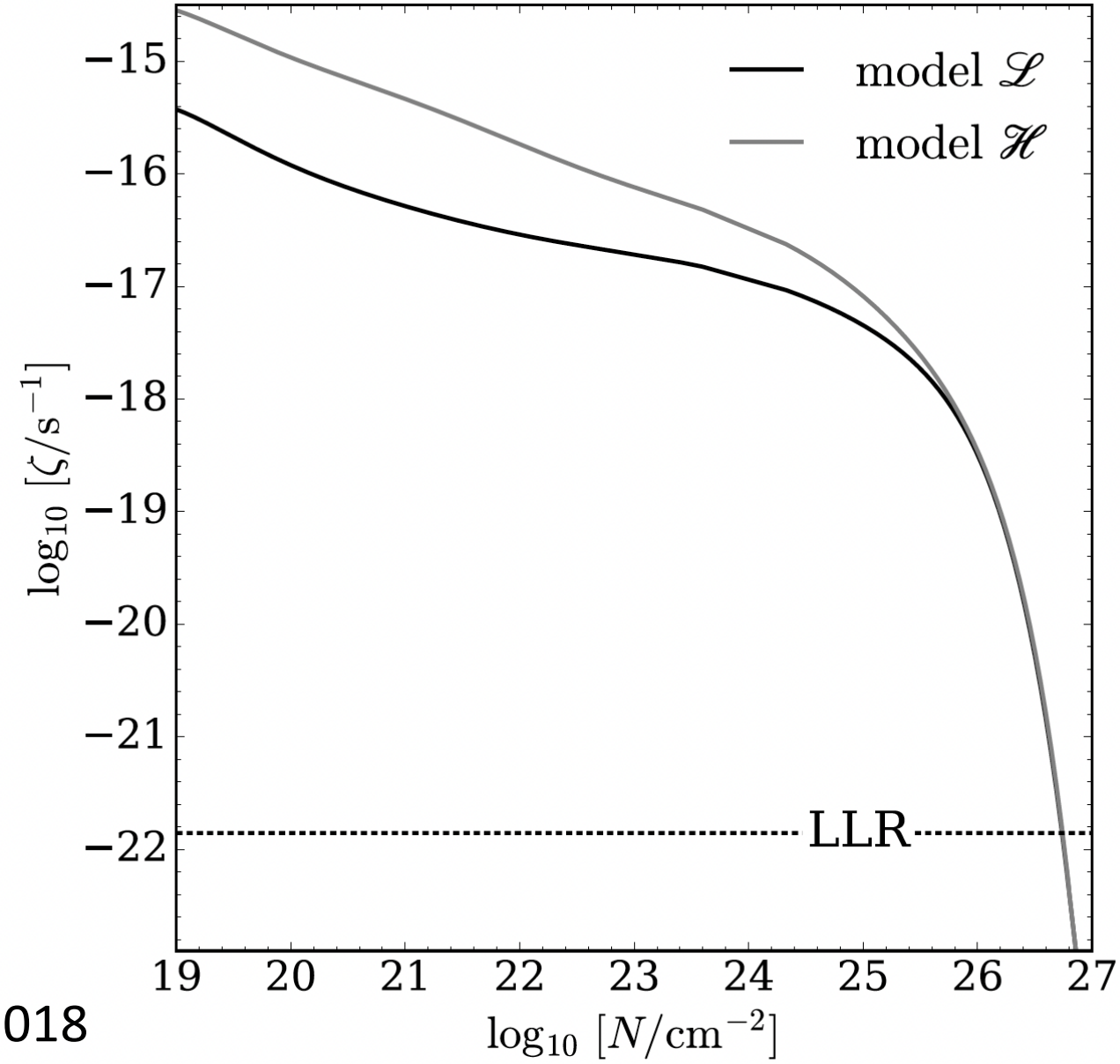


Enhanced CR flux



The effect is significant toward dense gas where the external ISRF is attenuated

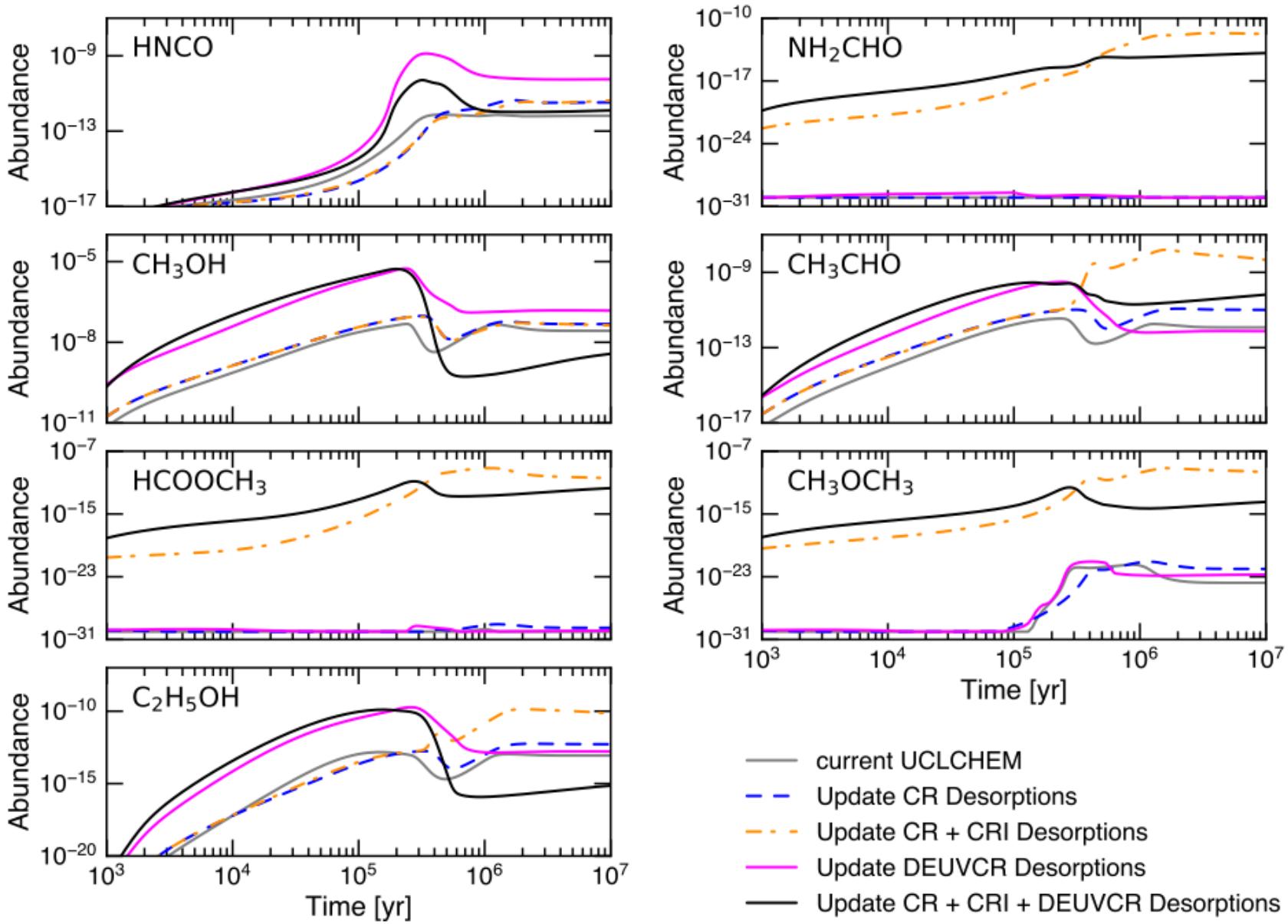
Cosmic-ray Ionisation Rate



Padovani et al. 2018

ALL to UCLCHEM

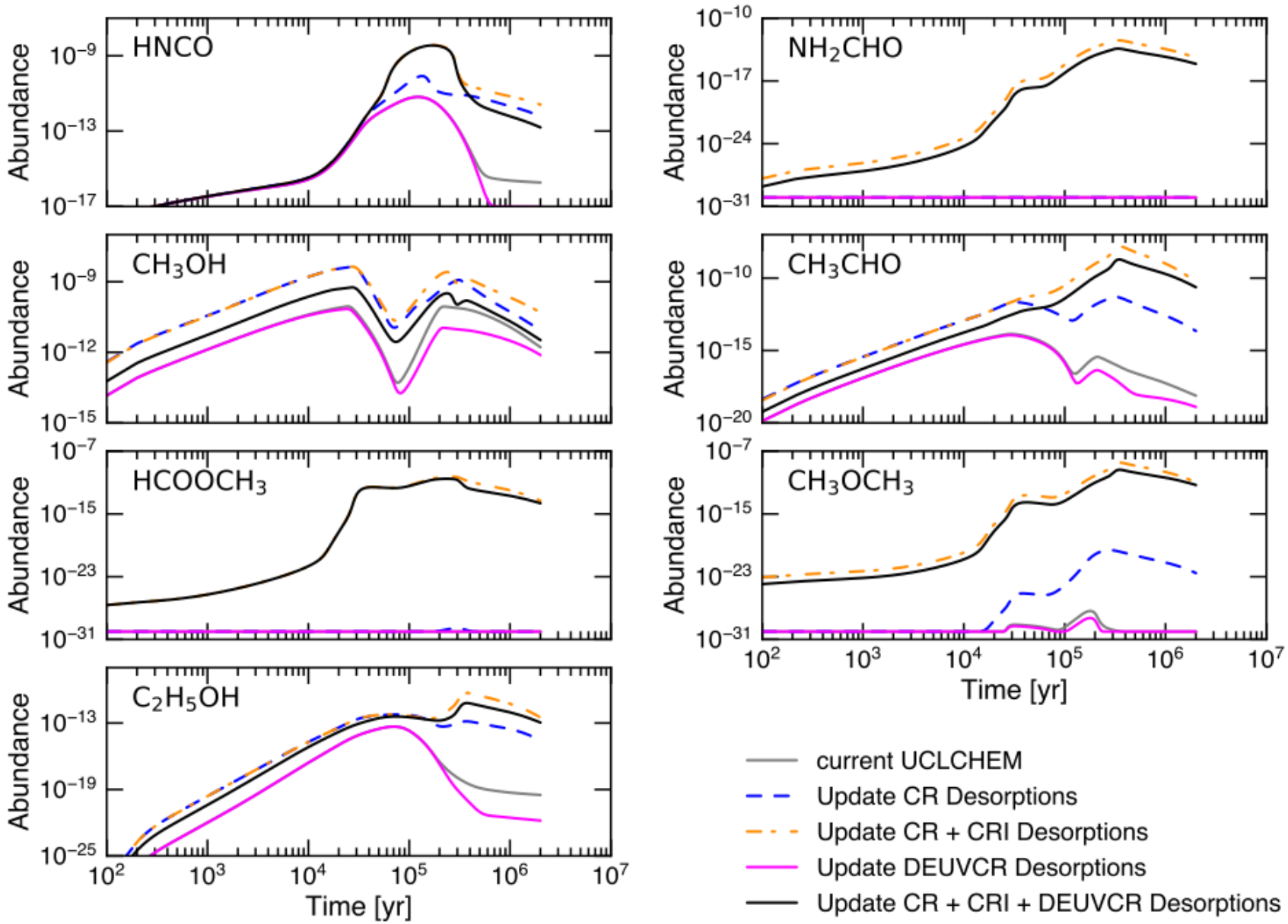
All updates to UCLCHEM



heating/cooling: OFF

Tram et al. in prep.

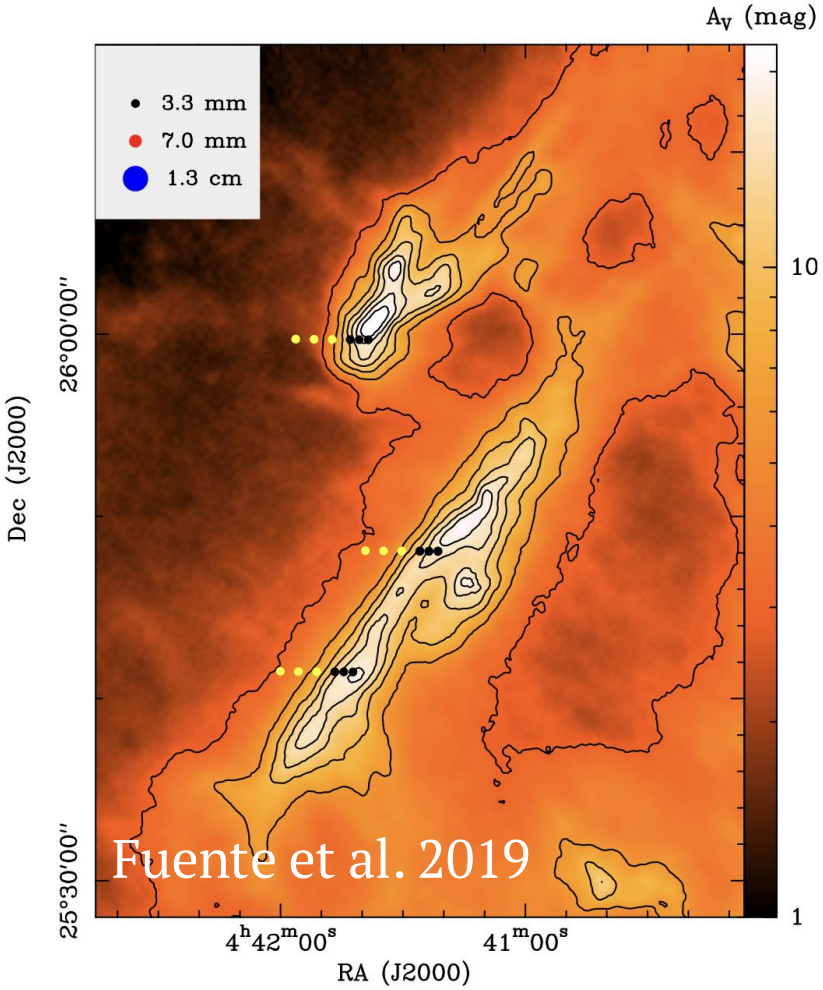
All updates to UCLCHEM



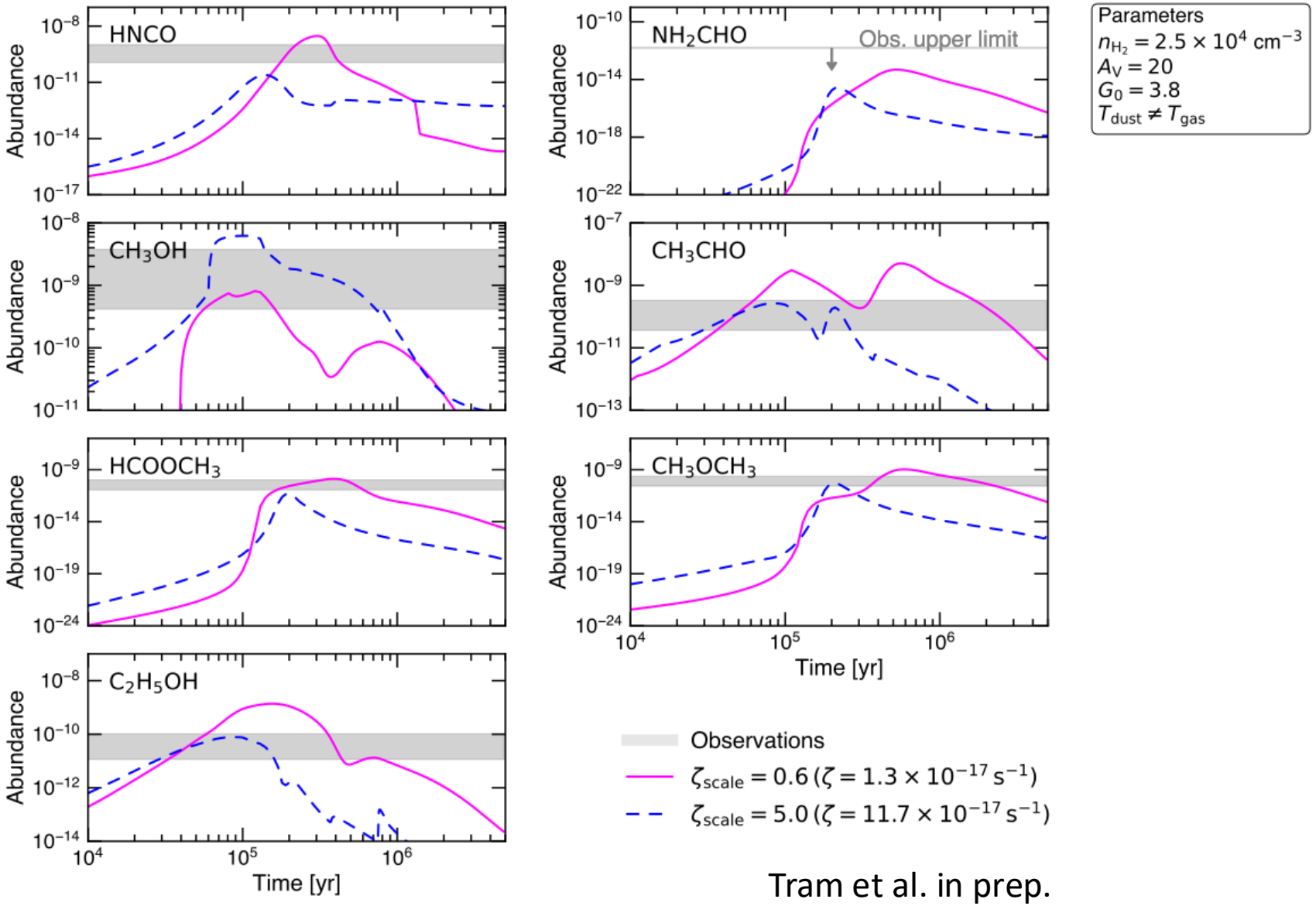
heating/cooling: OFF

Tram et al. in prep.

Application to TMC-1 Dark Cloud

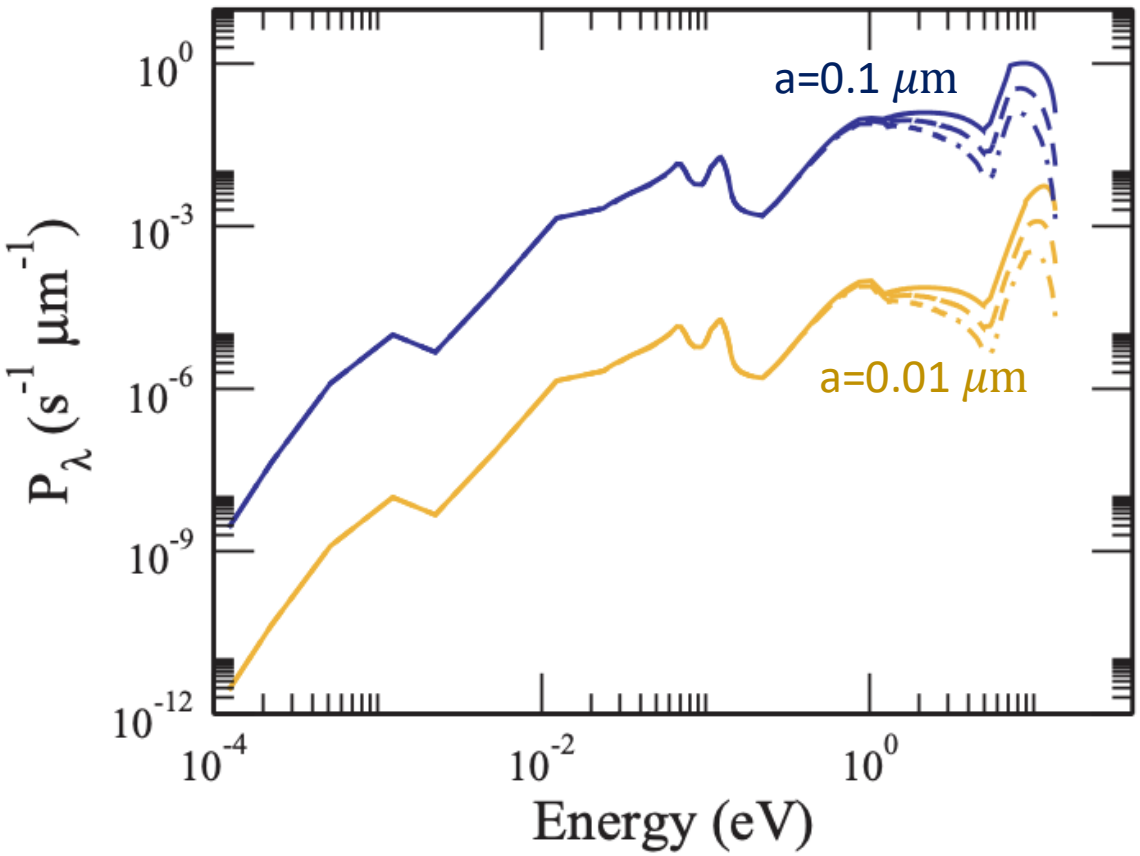


Fuente et al. 2019



Tram et al. in prep.

UV photon as a Quantized Particle



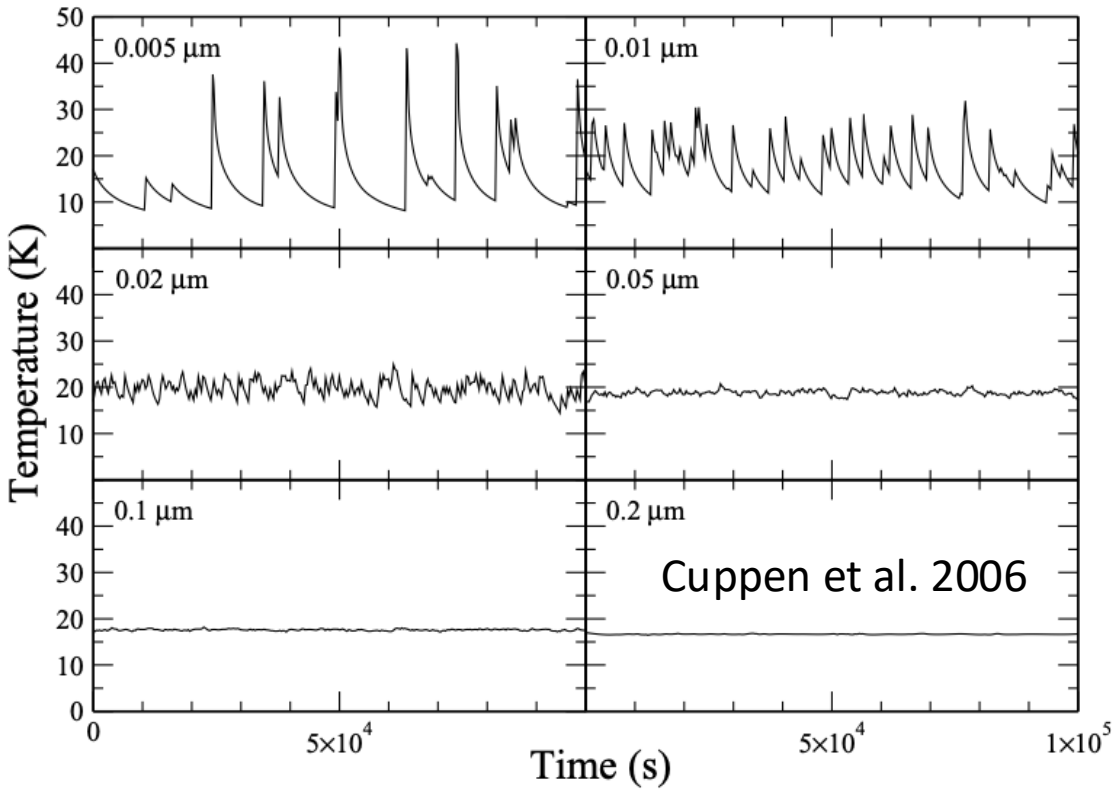
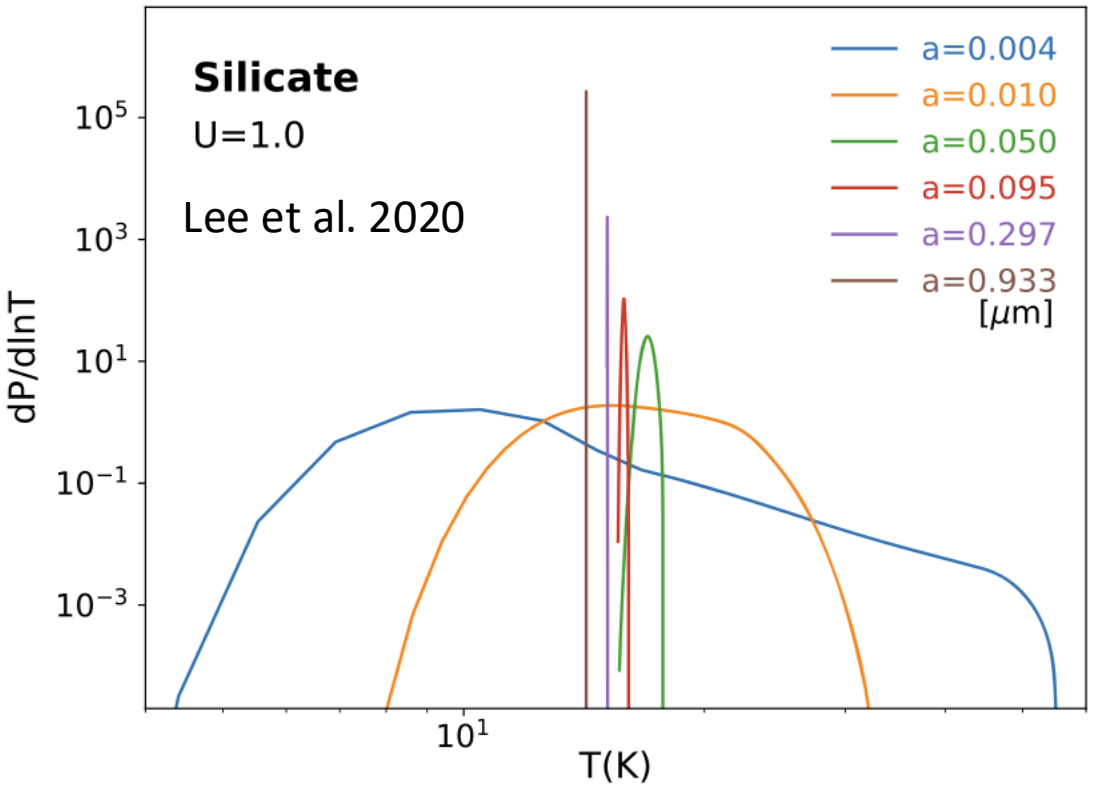
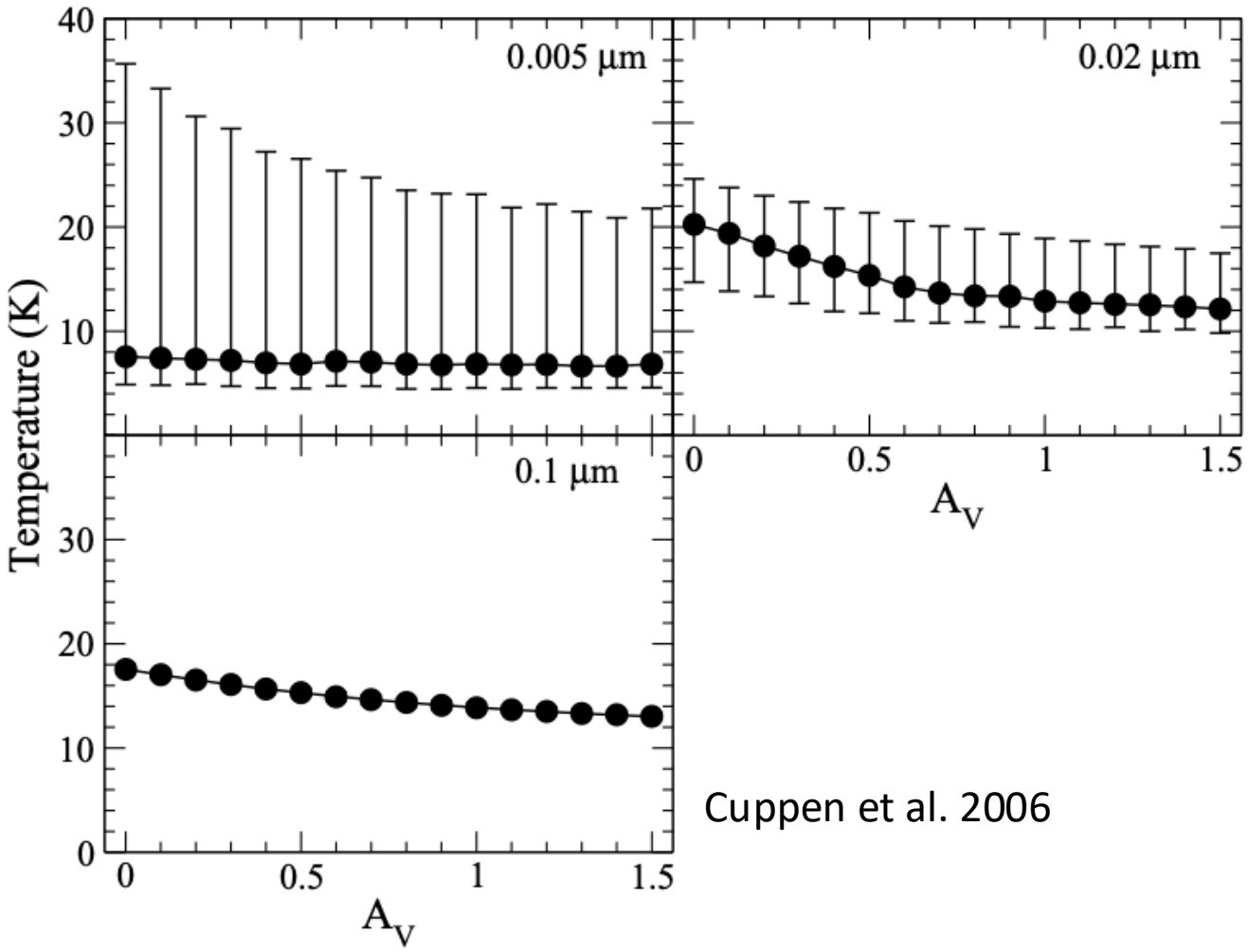


Figure 3. The temperature fluctuations as a function of time for different grain sizes and $A_V = 0$.

UV photon as a Quantized Particle



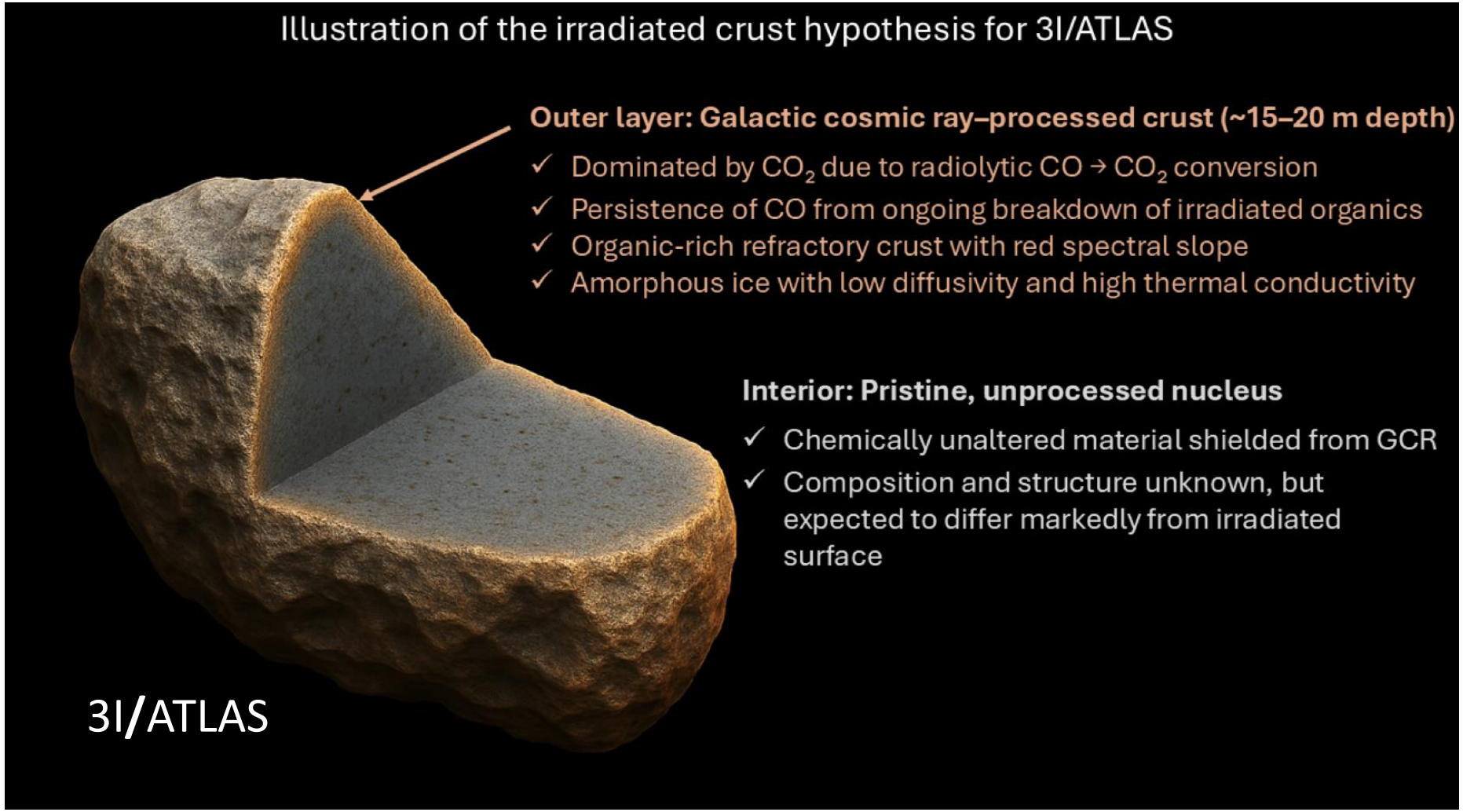


Cuppen et al. 2006

What to be Improved?

1. T_{CR} spectra is pre-computed → on-the-fly modelling
2. No CR sputtering on the grain
3. How dose the charge effect on the surface chemistry?
4. Electron can be captured back to grain → what happened next?
5. **<u>Effects of X-ray vs. Cosmic-ray?</u>**

Interstellar Comet vs. Cosmic-ray ?



Maggiolo et al. 2026
Roth et al. 2026

Extreme CO_2, CH_3OH

Thank you very much for your attention!